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# Trace elements in sulfide minerals from the Huangshaping copper-polymetallic deposit, Hunan, China: Ore genesis and element occurrence

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ABSTRACT

The Huangshaping copper-polymetallic deposit, which is located in southern Hunan Province, China, is an important resource base for Pb–Zn minerals. This study used the major sulfides (sphalerite, pyrite, pyrrhotite), which formed during the Pb–Zn mineralization stage of the Huangshaping deposit, to determine the genesis of the Pb–Zn ore minerals and element occurrence. In-situ LA-ICP-MS analysis of minerals was conducted to determine their trace element concentrations and reveal the distribution and occurrence of trace elements in different sulfides. The formation temperature and genesis of the Pb–Zn ore minerals were also constrained based on the deposit geology and trace element data. Trace elements were significantly different among different sulfides in the Pb–Zn mineralization stage of the deposit. Sphalerite primarily enriches Mn, Cu, Ag, Sn, and Pb and is the main carrier mineral of Ga, In, and Cd. Pyrite relatively enriches Tl, Sb, W, and Se, and pyrrhotite primarily enriches Ge. In sphalerite, Fe, Mn, Cd, Ga, In, Cu, Sn, Ag, and Ge exist by substituting for other elements (such as Zn). The replacement method of Ge is $Ge^{4+}+2Cu^{+}\leftrightarrow 3Zn^{2+}$, whereas that of In (Ga) is $In^{3+}(Ga^{3+}) + Cu^{+}\leftrightarrow 2Zn^{2+}$. In pyrrhotite, Ge is incorporated through coupling substitution, and the abnormal concentrations of Pb, Ag, and Se are caused by galena that has replaced pyrrhotite. In pyrite, W occurs as a solid solution. When galena replaces pyrite, it causes abnormal concentrations of Pb, Tl, Sb, and Ag. The sphalerite is characterized by high concentrations of high-temperature ore-forming elements (e.g., Fe, Mn, In, and Sn) and relatively low concentrations of Cd, Ge, Ga, and other elements. Thus, the sphalerite was formed at a medium–high temperature (288–341 °C). The trace element characteristics of sphalerite are consistent with those of high-temperature hydrothermal deposits. The deposit geology, pyrite Co/Ni ratio (averaging 0.13, n = 29), S/Se ratio (averaging 173,905, n = 19), pyrrhotite Co/Ni ratio (averaging 0.10, n = 26), and binary diagram of trace elements in sphalerite indicate that Pb–Zn mineralization is related to the late-stage hydrothermal activity of skarn mineralization. The Pb–Zn mineralization and W–Mo(Fe) mineralization together constitute a skarn-type mineralization system.

## 1. Introduction

The South China Metallogenic Province supplies many metals (e.g., Pb, Zn, Cu, Mo, Fe, W, and Sn) and is a significant resource producing area in China. Southern Hunan is an important part of this metallogenic province, with frequent magmatic and tectonic activities and various mineral resources, represented by the large-scale Huangshaping copper-polymetallic deposit. Many studies have been conducted on various aspects of this deposit, including zircon geochronology (Yao et al., 2005; Li et al., 2014a; Ai, 2013; Jiang et al., 2020), mineralization age and isotope geochemistry (Hua and Yao, 2006; Ma et al., 2007; Yao et al., 2007; Lei et al., 2010; Li et al., 2017), ore-forming fluid (Huang et al., 2013; Li et al., 2016; Li et al., 2019c), petrogenesis and geochemistry (Liu et al., 2009; Zhu et al., 2012; Li et al., 2014b; Hu et al., 2017), and the genesis of mineral deposits and sources of ore-forming materials (Ding et al., 2018a, b; Jiang et al., 2018). At Huangshaping copper-

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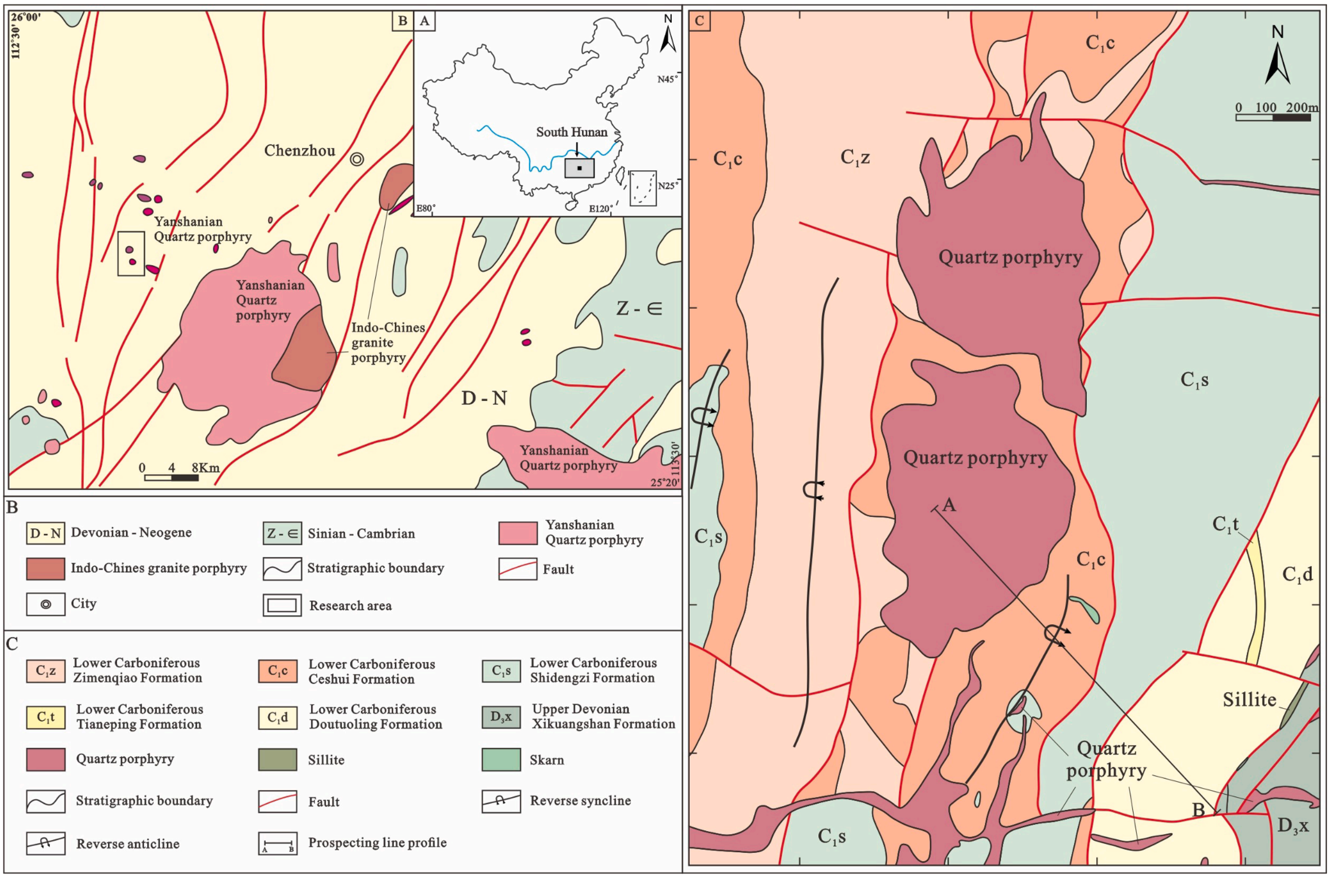


**Fig. 1.** A - Location of the study area; B - Regional geological map of the Pingbao area; C - Geological map of the Huangshaping deposit (modified after Zhu et al., 2021).

polymetallic deposit, there are two different types of ore minerals: the W–Mo(Fe) ore minerals and Pb–Zn ore minerals. The W–Mo(Fe) ore minerals have been characterized as a skarn-type mineralization (He et al., 2010; Ding et al., 2017; Zhao et al., 2018); however, the Pb–Zn mineralization has garnered less attention and remains unclear (Li et al., 2019b). Trace elements in sphalerite can provide valuable genetic information and are typically used to classify the genetic types of mineral deposits (Zhang, 1987; Ye et al., 2011; Liu et al., 2012; Wang et al., 2020; Li et al., 2021a, Li et al., 2021b; Zhang et al., 2021).

Currently, laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) in-situ microanalysis is widely used to accurately determine sulfide elemental compositions and forms (Cook et al., 2009; Ye et al., 2011). Few studies have used this method in the Huangshaping deposit (Ding et al., 2021). Thus, this method can be used to study the distributions and forms of trace elements in the Huangshaping deposit to fill gaps in this field. Sphalerite, pyrite, and pyrrhotite, which formed during the Pb–Zn mineralization stage, are the main minerals of the Huangshaping deposit. These mineral samples were subjected to in-situ LA-ICP-MS analysis to (i) identify the internal geochemical characteristics of different sulfide compounds, (ii) determine the sulfide mineralization temperature, and (iii) constrain the genesis of the Pb–Zn orebody in the Huangshaping copper-polymetallic deposit.

## 2. Geological setting of the Huangshaping deposit

Southern Hunan, located at the central section of Nanling orogenic belt, is an important polymetallic area in the metallogenic provinces of South China, encompassing the Dongpo, Shizhuyuan, Xintianling, Furong, Xianghualing, Yaogangxian, Baiyunxian, and Baoshan deposits (Mao et al., 2021; Xie et al., 2021), as well as the Huangshaping copper-polymetallic deposit (Fig. 1-a,b). The Nanling orogenic belt is characterized by active tectonism, strong tectonic deformation, multiple tectonic cycles, and multiple periods and episodes of fault activity. The Huangshaping deposit is located on the NW margin of the central section of Nanling orogenic belt, between the Pingxiang–Guilin and Ganjiang fault zones (Fig. 1-b). Four groups of tectonic structures can be identified in the Huangshaping deposit; the directions of the main structures are NE–NNE, with secondary structures exhibiting NN, EW, and NW. In this area, the Yanshan orogeny is the tectonic event most closely associated with mineralization in temporal and spatial terms (Shu et al., 2006; Zhu et al., 2010; Zhang et al., 2013; Zhao et al., 2014). The multiple tectonic movements in this area have yielded many deep and extensive faults, regional faults, and fold systems. These structures provide a suitable setting for magma intrusion and eruption, which are beneficial for magma transportation and storage (Zhu et al., 2010; Han et al., 2020; Tao et al., 2020).

The main lithostratigraphic units exposed in the Huangshaping deposit include the Lower Carboniferous Zimenqiao Formation ($C_1z$), Ceshui Formation ($C_1c$), and Shidengzi Formation ($C_1s$) (Fig. 1-c). The $C_1s$ limestone is the main ore-bearing rock and can be further divided into three types based on lithology and alteration degree (from the bottom up): (i) gray medium-thick laminated limestone with weak calcite alteration, a small amount of dolomite, thin laminated argillaceous limestone, and carbonaceous limestone, and no mineralization; (ii) gray medium-thick layered limestone with strong calcite alteration, in which calcite occurs as nodules, irregular, or veinous and develops mineralization (pyrite, chalcopyrite veinlets); and (iii) dark gray medium-thick laminated limestone, without alteration. The Ceshui Formation ($C_1c$) can be divided into (i) yellow-white to gray thin-medium calcareous and argillaceous sandstone exhibiting a fine-

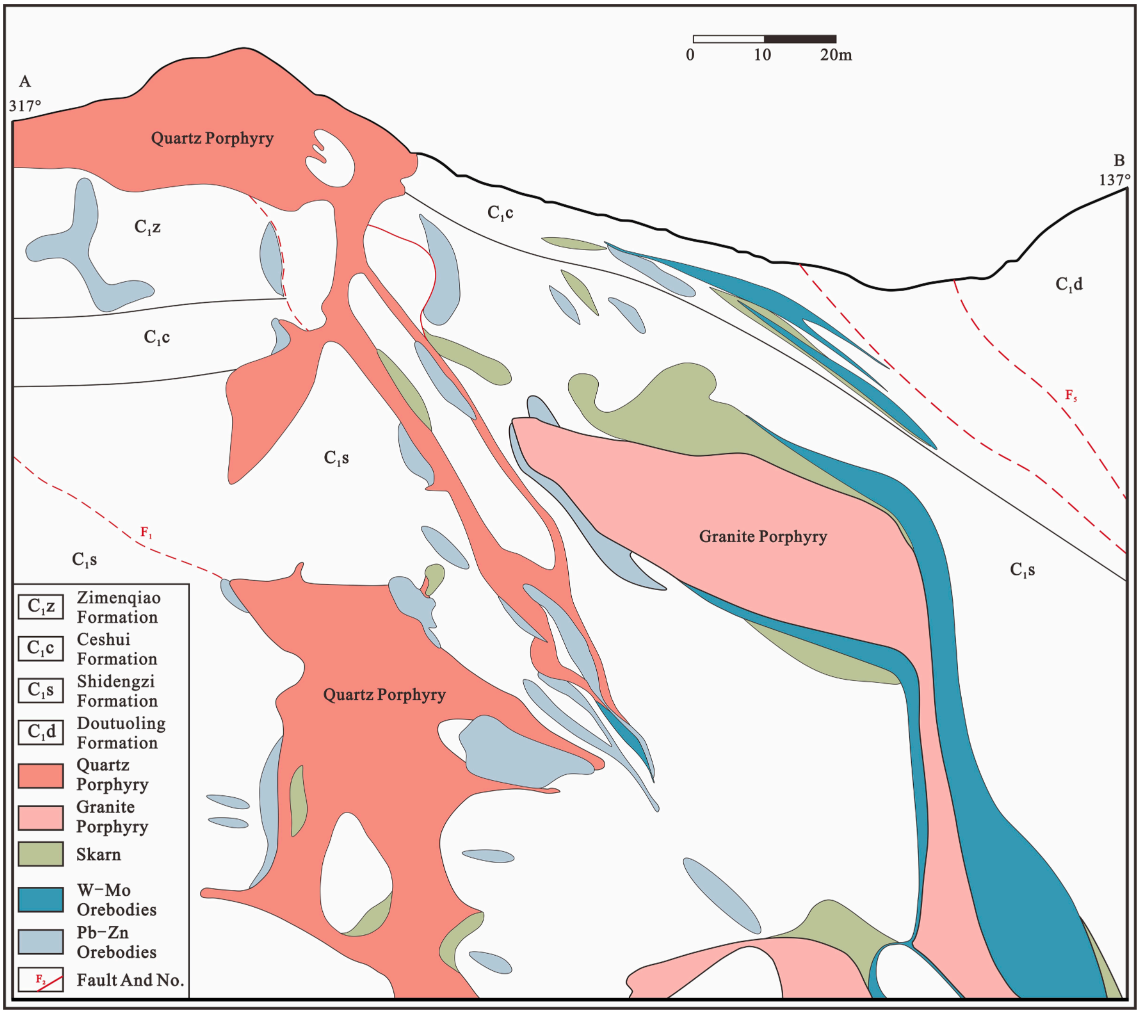


**Fig. 2.** Section view of exploration line No. 105 in the Huangshaping deposit mining area.

grained structure and joint, (ii) thin layered argillaceous sandstone interbedded with black carbonaceous shale, and (iii) gray-white thin layered quartz sandstone with a fine-grained structure, joints, and broken rocks, with quartz veins. The Zimenqiao Formation ($C_1z$) is grayish-white course-medium-grained thick laminated massive dolomite, developing calcite and pyrite and chert mass; calcite occurs as nodules and veins, whereas pyrite occurs as veins.

The Huangshaping copper-polymetallic deposit is closely associated with the magmatic rocks both spatially and temporally. The quartz porphyry, monzonite granite porphyry, and dacite porphyry are exposed in the orefield, and their zircon U–Pb ages are 160.8 ± 1.0 Ma, 155.2 ± 0.4 Ma, and 158.5 ± 0.9 Ma, respectively (Yuan et al., 2014). The Rb–Sr age of sphalerite, which is from the Huangshaping deposit, is 143 ± 1 Ma (Lu et al., 2015), and the Re–Os age of molybdenite is 150.9–156.9 Ma (Yao et al., 2007). Therefore, the age ranges of the magmatic rocks are consistent with the age ranges of Huangshaping deposit mineralization, indicating that both formed through large-scale magmatism and mineralization in southern Hunan during the Yanshan orogeny (Lei et al., 2010; Quan et al., 2012; Yuan et al., 2014; Yuan et al., 2018). The magmatic rocks in the orefield are small and numerous (Fig. 1-b,c); most occur as dykes, sills, and stocks (Fig. 2). The spatial distribution of individual intrusions is strictly controlled by the fault structure (He et al., 2010; Han et al., 2020). These intrusions were emplaced in the limestone, while the wall rocks developed complex types of alterations, including skarn (garnet, diopside, fluorite, epidote, chlorite), and calcite (Zhao et al., 2016a; Zhao et al., 2016b; Tao et al., 2020; Zhu et al., 2021). The main skarn minerals are garnet, diopside, fluorite, chlorite, and epidote. The skarn is controlled by intrusions (quartz porphyry and granite porphyry) and wall rocks ($C_1s$ limestone) and is closely related to mineralization. Almost all the ore minerals were developed in the contact zones between the intrusions and limestone (Fig. 2, Fig. 3).

Most orebodies develop in skarn, weak skarn alteration limestone, and wall rocks (Zhu et al., 2010; Ding et al., 2017; Han et al., 2020; Tao et al., 2020). The shapes and boundaries of most of the orebodies (all the W–Mo(Fe) orebodies and some of the Pb–Zn orebodies) are consistent with those of skarns and intrusions (quartz and granite porphyries). Some orebodies (primarily Pb–Zn orebodies) occur in wall rocks ($C_1s$ limestone) adjacent to skarn (Fig. 2, Fig. 3). The boundaries of Pb–Zn orebodies are beyond the skarn boundaries, and the Pb–Zn orebodies occur in distal wall rock, indicating that the Pb–Zn mineralization may have occurred after the skarn formation. These orebodies primarily occur as veinous, columnar, lenticular, cystic, and lamellar features (Fig. 4-a,b,c). From the center outward, the deposit zones can be divided based on mineralization and alteration into intrusion, anhydrous skarn zone, hydrous skarn zone, hydrous skarn with or without a W–Mo(Fe) mineralization zone, and Pb–Zn mineralization zone (Zhao et al., 2016b; Han et al., 2020; Tao et al., 2020).

The main ore minerals of the W–Mo(Fe) mineralization are scheelite and magnetite, and the main gangue minerals include garnet, diopside, tremolite, epidote, and fluorite. The alteration in the wall rocks is skarn (primarily hydrous skarn including tremolite and diopside skarns). The main ore minerals of the Pb–Zn mineralization are sphalerite, galena,

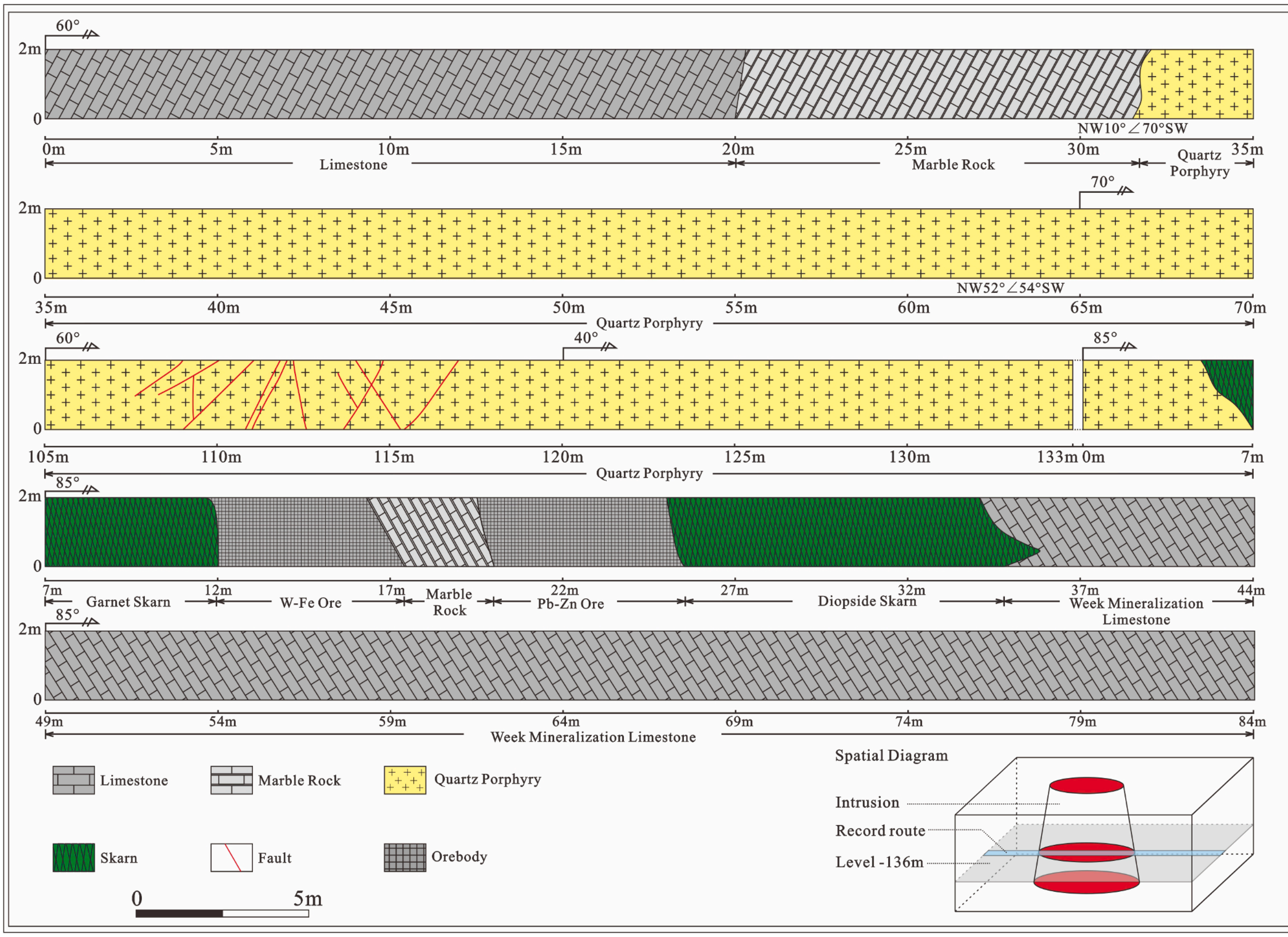


**Fig. 3.** Level −136 m mining tunnel geological record of the Huangshaping deposit.

pyrite, and chalcopyrite, and the main gangue minerals are calcite and fluorite. Additionally, the alterations in the wall rocks are primarily fluorite, chlorite, and calcite alterations (Fig. 5-a,b,c).

The Huangshaping deposit mineralization can be divided into three mineralization periods and six mineralization stages (Fig. 6). The periods comprised sedimentary, skarn, and sulfide mineralizations. The skarn and sulfide periods are the main mineralization periods of the Huangshaping deposit. Magnetite was formed during stage III (hydrous skarn stage), and cassiterite and scheelite were formed during stage IV (oxide stage). Typical high–medium-temperature sulfides such as molybdenite, bismuthate, bornite, chalcopyrite, and pyrite were primarily formed during stage V (high–medium-temperature sulfide stage) and stage VI (medium–low-temperature sulfide stage), which featured much of the high-temperature ore mineral generation. Sphalerite and galena were formed during stages V and VI. Carbonate minerals, such as calcite and siderite, were formed during stage VI. The main stages of W–Mo(Fe) mineralization are stages III, IV, and V, whereas those of Pb–Zn mineralization are stages V and VI (Fig. 6).

## 3. Samples and analytical methods

The Pb–Zn ore minerals were collected from mine tunnels at −96 m, −136 m, −156 m, −176 m, and −216 m in the Huangshaping deposit. Over 60 polished thin sections were prepared and analyzed via transmitted and reflected light microscopy; ten representative samples were analyzed via back-scattered electron (BSE) and LA-ICP-MS to identify minerals, examine mineral microstructure, and determine compositions. Samples containing pyrite, pyrrhotite, and sphalerite were selected from the main stage of Pb–Zn mineralization in the deposit. Their formation sequence was identified as pyrite → pyrrhotite → sphalerite.

Petrographic observation of these minerals were carried out using a Nova NanoSEM 450 field emission scanning electron microscope (FSEM), at the State Key Laboratory of Lithospheric Evolution, Institute of Geology and Geophysics, Chinses Academy of Sciences. Backscattered electron (BSE) images were acquired with an accelerating voltage of 15 kV and a primary beam current of 20nA. The instrument is equipped with an X-MAXN80 Energy-Dispersive X-ray Spectrometer (EDS) detector, which is used for semiquantitative spot analyses and elemental mapping. BSE imaging was used to characterize the compositional heterogeneity (e.g., zoning) and mineral inclusions, which would have affected the trace element distributions (Fig. 5-a, b, c, d).

Trace element analyses of sulfides were conducted via LA-ICP-MS at the State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences. Laser sampling was performed using an ASI RESOLution-LR-S155 laser microprobe equipped with a Coherent Compex-Pro 193 nm ArF excimer laser. An Agilent 7700x ICP-MS instrument was used to acquire ion-signal intensities. The ablated aerosol was mixed with Ar (900 ml/min) as a transport gas, before exiting the cell. The following isotopes were monitored: $^{25}Mg$, $^{27}Al$, $^{29}Si$, $^{34}S$, $^{43}Ca$, $^{45}Sc$, $^{49}Ti$, $^{51}V$, $^{53}Cr$, $^{55}Mn$, $^{57}Fe$, $^{59}Co$, $^{60}Ni$, $^{65}Cu$, $^{66}Zn$, $^{71}Ga$, $^{72}Ge$, $^{75}As$, $^{77}Se$, $^{93}Nb$, $^{95}Mo$, $^{107}Ag$, $^{111}Cd$, $^{115}In$, $^{118}Sn$, $^{121}Sb$, $^{125}Te$, $^{139}La$, $^{181}Ta$, $^{184}W$, $^{197}Au$, $^{205}Tl$, $^{208}Pb$, and $^{209}Bi$. We used a consistent 26 μm-diameter spot size for all the analyses. Each analysis involved a background acquisition of approximately 30 s (gas blank) followed by 60 s of data acquisition from the sample. STDGL3 was used to determine the chalcophile and siderophile concentrations

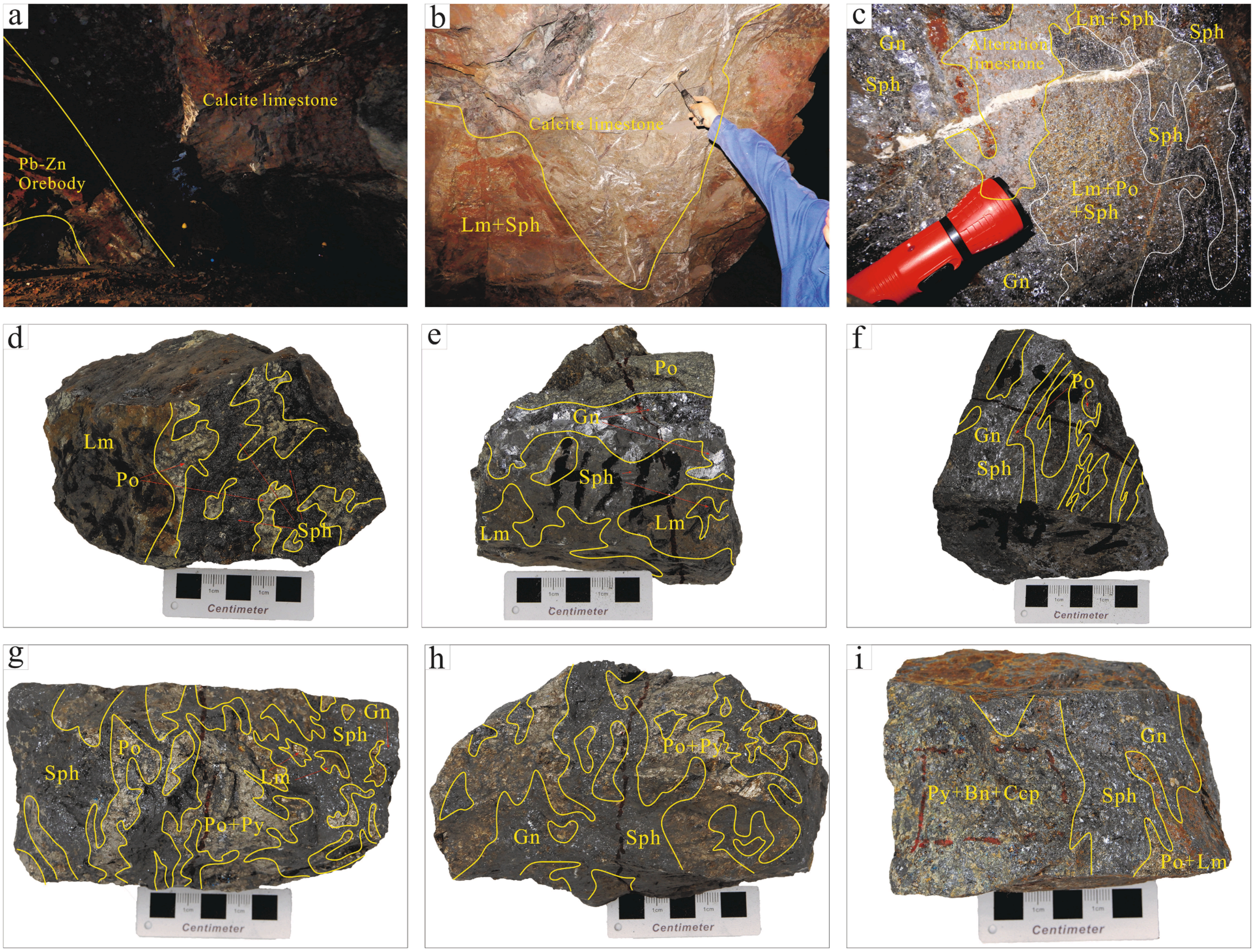


**Fig. 4.** a - Photo of the mined-out orebody area; b, c - Photo of tunnel orebody; d - HSP1; e - HSP2; f - HSP3; g, h - HSP8; I - HLR68. Minerals occur as veins and disseminate, the early minerals (Bn, Py, Po) have been replaced by the late minerals (Sph, Gn), and Py and Po have been oxidized to form Lm; Lm - limonite, Po - pyrrhotite, Sph - sphalerite, Gn - galena, Py - pyrite, Bn - bornite.

(Danyushevsky et al., 2011), and the in-house standard of pure pyrite Py was used to calibrate the S and Fe concentrations. The integrated count data of lithophile element concentrations were calibrated and converted using GSD-1G. Sulfide reference material, MASS-1, was analyzed as an unknown sample to check the analytical accuracy.

## 4. Mineralogy and analysis results

### 4.1. Mineralogy

The sulfides occur as veins and massive ores (Fig. 4) and have granular crystals or replacement textures. For example, the subhedral–euhedral crystals, pyrite and arsenopyrite, and subhedral-xenomorphic pyrrhotite have been replaced by the xenomorphic-subhedral crystals galena and sphalerite. Pyrrhotite occurs as relict replacement, relict islands, or subhedral-xenomorphic crystals textures. Pyrite and arsenopyrite occur as relict islands in pyrrhotite or sphalerite. There are galena micro-inclusions in sphalerite, implying sphalerite replacement by late galena (Fig. 5). The pyrite, pyrrhotite, and sphalerite occur with large grain crystals of fluorite and weak skarn alteration minerals (epidote and chlorite). In some samples, the galena was observed to associate with the mineral that enriches Pb, Sb, and Sn. All the sulfides have been cut (and locally destroyed) by calcite (enrich Fe) and/or siderite veins.

Sphalerite was formed during stage V and is mostly associated with galena (Fig. 4-f,g,h,i; Fig. 5-a,b,f,l), which represents late-stage mineral assemblages. Most sphalerites have subhedral-xenomorphic crystalline textures and occur as vein or mass. When sphalerite replaces pyrrhotite, pyrrhotite has the replaced relict texture. Sphalerite contains other minerals of relict islands, such as pyrrhotite and other early minerals, as well as late stage fine-grain galena.

The pyrite and arsenopyrite assemblage was formed in stage V (Fig. 5-a,d,e,i,j,k). Pyrite and arsenopyrite are primarily euhedral–subhedral crystals that have been replaced by late minerals (e.g., pyrrhotite and sphalerite) and are distributed within the late minerals as relict islands. Xenomorphic-subhedral crystal pyrite (Py(L); Fig. 5-f) was also identified but represents late-stage pyrite, which was not examined in this study.

Pyrrhotite, formed during the medium stage V, occurs as veins and disseminated structures and is characteristic by the subhedral-xenomorphic crystals (Fig. 4-d,f,g,h; Fig. 5-a,g,i,j,k,l). Pyrrhotite has replaced pyrite and arsenopyrite and has been replaced by late minerals (e.g., sphalerite and galena) to form replaced relict or relict islands textures.

### 4.2. Analysis results

Trace element composition data of sphalerite (10 samples, 78 spots), pyrite (6 samples, 38 spots), and pyrrhotite (5 samples, 41 spots) from the Huangshaping deposit are summarized in Table 1. Representative

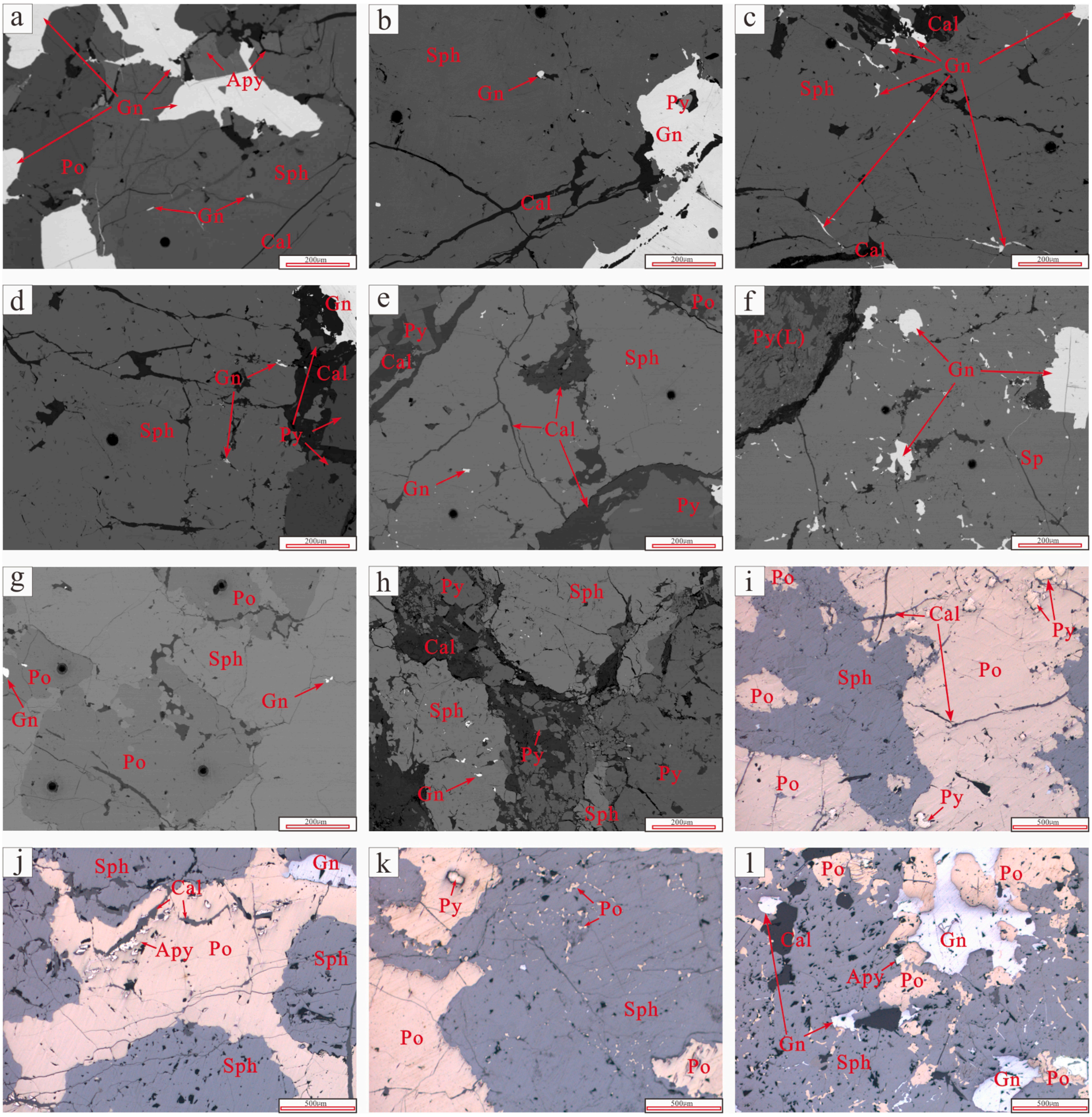


**Fig. 5.** a, b, c, d, e, f, g, h - BSE photographs of HSP5, HSP6, HSP6, HSP6, HSP7, HSP8, HSP10, and HSP12, respectively; i, j, k, l - Microscopic photographs of HSP10, HSP9, HSP9, and HSP5, respectively; a, b, d, e, h, l: Late Gn fine-grain occurs in Sph; d: Late Gn fine-grain occurs in Sph, and Py has been replaced by Gn; f: Gn has replaced Sph, Py(L) is xenomorphic-subhedral crystals; g: Sph has replaced Po; i, j: Sph and Gn have replaced Po, replaced residual Py and Apy can be observed in Po; k: Sph has replaced Po, the virus-like Po can be observed in Sph, and the replaced residual Py can be observed in the Po; Py - pyrite; Po - pyrrhotite; Apy - arsenopyrite; Sph - sphalerite; Gn – galena; Cal - calcite.

time-resolved LA-ICP-MS signal depth profiles are shown in Fig. 9. The Complete LA-ICP-MS datasets are provided in the Electronic Appendix A (Supplemental Material 1).

### 4.3. Sphalerite

The sphalerite samples exhibited significant Fe content (8.75 wt%–14.6 wt%, averaging 11.8 wt%); most of these can be described as ferrosphalerite (Fe greater than 10 wt%). The Cu and Mn contents exhibited wide ranges (54.7–35,121 ppm, averaging 2,125 ppm and 370–2329 ppm, averaging 1,336 ppm, respectively). The Cd concentration in sphalerite had a relatively narrow range (1,908–3,966 ppm, averaging 2,915 ppm). Although Pb, Sn, In, and Ag are high-content trace elements in sphalerite, they exhibited wide ranges (1.89–15,206 ppm, averaging 417 ppm; 0.39–4,365 ppm, averaging 215 ppm; 0.19–425 ppm, averaging 64.4 ppm; and 7.69–322 ppm, averaging 57.0 ppm, respectively).

Ga, Ti, Sb, and Ge had relatively lower contents ranging 2.16–58.5 ppm (averaging 12.1 ppm), <DL to 103 ppm (averaging 7.12 ppm), <DL to 27.6 ppm (averaging 3.65 ppm), and < DL to 4.01 ppm (averaging 2.56 ppm). Among the trace elements analyzed in this study, Se, Cr, Bi, V, W, Te, As, Co, Tl, Ni, Sc, Mo, La, Au, Nb, and Ta with contents above the DL or averaging concentrations < 1 ppm were detected in a few areas.

### 4.4. Pyrite

Mn was the highest concentration trace element in pyrite samples and exhibited a wide range (<DL to 1471 ppm, averaging 105 ppm). All the pyrite samples detected Pb, Ge, and Ti, with relative contents of

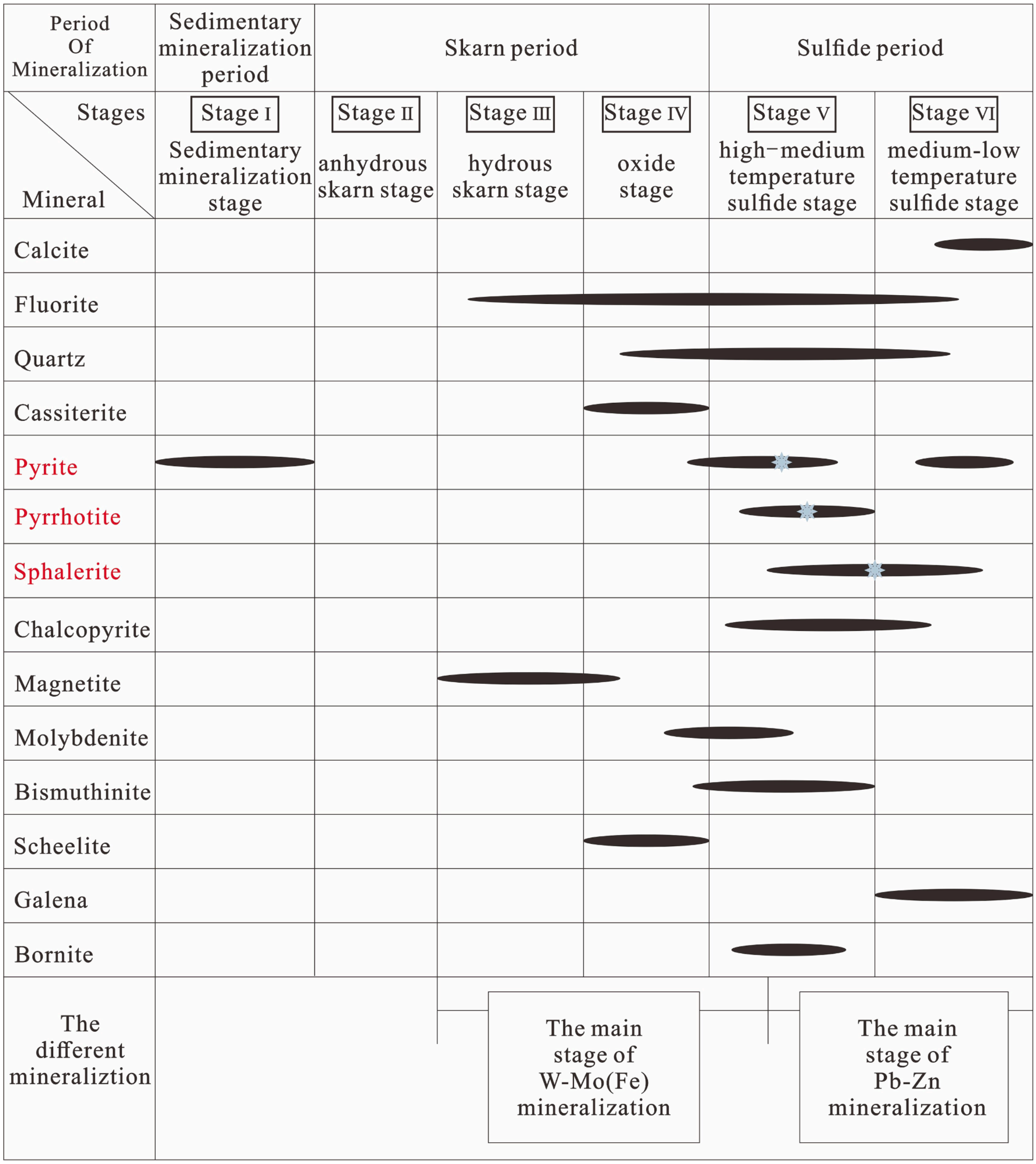


**Fig. 6.** Mineralization stages of the Huangshaping deposit.

0.53–93.9 ppm (averaging 19.114 ppm), 7.66–10.3 ppm (averaging 8.63 ppm), and 3.57–10.8 ppm (averaging 7.53 ppm), respectively. W and As had relatively higher contents, ranging < DL to 68.2 ppm (averaging 24.1 ppm) and < DL to 122 ppm (averaging 9.80 ppm). Sn, Ni, Tl, Sb, and Se are high-content elements in pyrite but exhibited wide ranges (<DL to 70.5 ppm, averaging 4.84 ppm; <DL to 39.9 ppm, averaging 7.54 ppm; <DL to 31.2 ppm, averaging 4.99 ppm; <DL to 26.7 ppm, averaging 5.41 ppm; and < DL to 20.9 ppm, averaging 5.78 ppm; respectively).

The Zn, Ag, Cu, and Co contents ranged from < DL to 9.17 ppm (averaging 2.58 ppm), <DL to 9.01 ppm (averaging 2.18), <DL to 4.78 ppm (averaging 1.42 ppm), and < DL to 1.52 ppm (averaging 0.58 ppm), respectively. Additionally, the Te, Bi, V, Cr, Sc, La, Cd, Mo, Ga, In, Au, Ta, and Nb contents in pyrite were mostly below the DL or maximum < 1 ppm.

### 4.5. *Pyrrhotite*

The Ge and Ti contents of all the pyrrhotite samples were above the DL and exhibited narrow ranges (8.17–14.4 ppm, averaging 11.0 ppm and 4.03–14.7 ppm, averaging 7.39 ppm, respectively). The Zn, Pb, and Se contents were higher but exhibited wide ranges (<DL to 512 ppm, averaging 95.4 ppm; <DL to 241 ppm, averaging 17.1 ppm; and < DL to 46.4 ppm, averaging 14.9 ppm, respectively). The Ni and Ag contents in the pyrrhotite samples were low and exhibited wide ranges (<DL to 21.9 ppm, averaging 3.02 ppm and < DL to 11.6 ppm, averaging 3.77 ppm, respectively).

Cu, Mn, Cd, Sn, In, Bi, W, Cr, Sb, As, Te, Sc, Tl, V, Ga, Co, Mo, Au, La, Ta, and Nb were detected in a few spots with contents above the DL or average concentrations < 1 ppm.

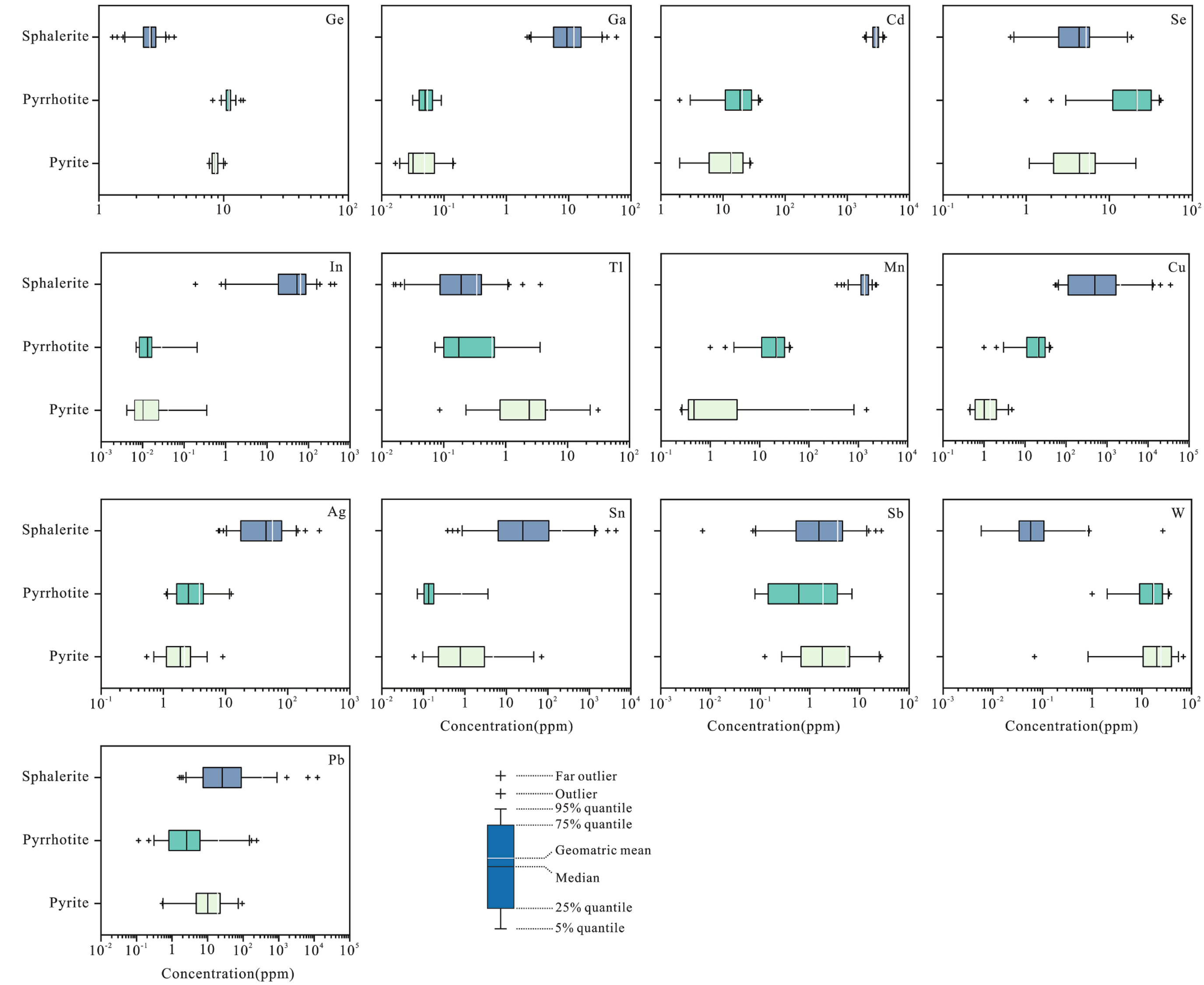


**Fig. 7.** Comparison of trace and dispersed elements in sphalerite, pyrrhotite, and pyrite in the Huangshaping deposit.

## 5. Discussion

### *5.1. Trace element distributions in different sulfides*

Sphalerite, pyrite, and pyrrhotite contained various trace elements, and the concentrations of these elements varied among the different minerals. However, a few studies have been conducted on the element distribution in different sulfides (e.g. Ye et al., 2012; George et al., 2016; Knorsch et al., 2020; Ye et al., 2016). Fe, Mn, Cd, and Cu were enriched in sphalerite; Mn, W, and Pb were enriched in pyrite; and Ge, Zn, Pb, Se were enriched in pyrrhotite. Although some elements were enriched in pyrrhotite, their concentrations were lower than those in sphalerite and pyrite. In this study, the trace element distributions of sphalerite, pyrite, and pyrrhotite were similar to those observed in previous studies.

In, Cd, and Ga were enriched in sphalerite. The concentrations of In in sphalerite exhibited a wide range with a maximum of 425 ppm. The concentrations of In in pyrrhotite and pyrite were generally below the DL; a few spots (pyrrhotite: n = 14, pyrite: n = 19) were above the DL, with In contents of < 1 ppm. The In content in sphalerite was 2–3 orders of magnitude higher than those in pyrrhotite and pyrite. The In distribution can be summarized as $In_{Sph} > In_{Py}$ and $In_{Po}$. Cd content in sphalerite samples (1,908–3,966 ppm, averaging 2,915 ppm) exhibited a uniform distribution and was generally 3–4 orders of magnitude higher than that in pyrrhotite and pyrite. The Cd contents in pyrrhotite and pyrite were above the DL in only a few spots, and Cd contents in pyrrhotite and pyrite were < 1 ppm. The distribution of Cd was $Cd_{Sph} > Cd_{Po}$ and $Cd_{Py}$. Although the Ga concentration in pyrite and pyrrhotite was 2 orders of magnitude lower than that in sphalerite (Fig. 7), the Ga concentration was below DL in some pyrrhotite samples; however, in others, it was higher than that in pyrite. Although sphalerite was rich in Ga, the Ga concentration varied widely from 2.16 to 58.5 ppm and was not uniformly distributed. The Ga distribution was $Ga_{Sph} > Ga_{Po}$ and $Ga_{Py}$.

Although Ge was enriched in pyrrhotite, the difference in concentration was small and < 1 order of magnitude higher than that of sphalerite and pyrite. The Ge distribution was $Ge_{Po} > Ge_{Py} > Ge_{Sph}$. Se was enriched in pyrite, and the Se contents in sphalerite, pyrrhotite, and pyrite were above the DL in a few samples (sphalerite: n = 37, pyrite: n = 19, and pyrrhotite: n = 9). The Se distribution in different sulfides did not appear to be uniform and had a Se is $Se_{Py} > Se_{Po} > Se_{Sph}$ distribution. Tl was enriched in pyrite, and the Tl content in pyrite was 1 order of magnitude higher than that in sphalerite and pyrrhotite. The Tl concentration in pyrrhotite was above DL for only a few samples (pyrrhotite: n = 12), however, there was little difference in the Tl concentration

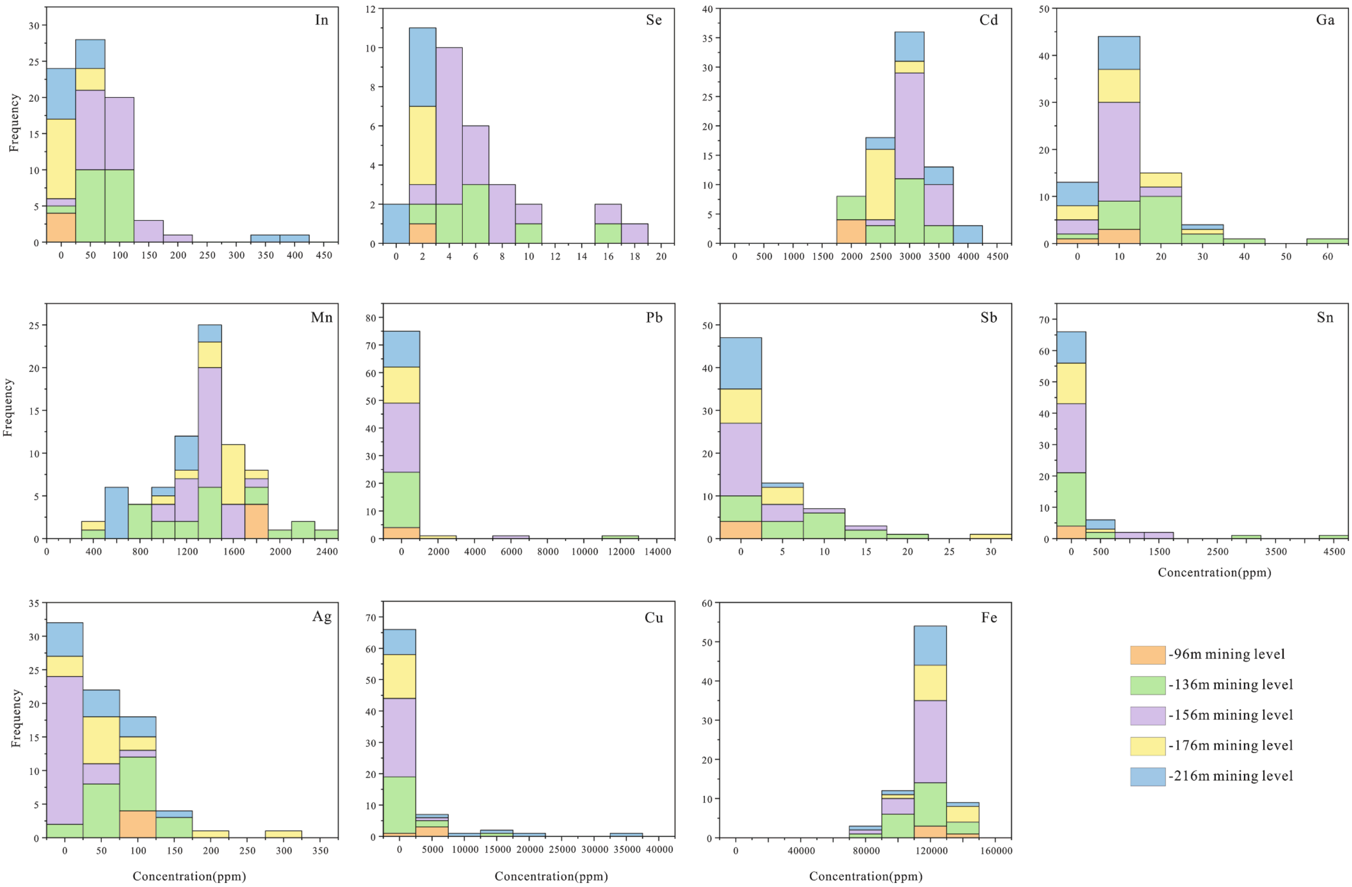


**Fig. 8.** Frequency histogram of the contents of sphalerite elements.

among the three sulfides. The Tl distribution was $Tl_{Py} > Tl_{Po} > Tl_{Sph}$.

Mn, Cu, Ag, Sn, and Pb were enriched in sphalerite. Mn content in sphalerite was high, and the Mn content in pyrite exhibited a wide range. Mn contents of most spots in pyrrhotite were below DL. The Mn distribution was $Mn_{Sph} > Mn_{Py} > Mn_{Po}$. The Cu content in sphalerite exhibited a wide range and was approximately 1 ppm in most of the pyrite and pyrrhotite. The distribution of Cu was $Cu_{Sph} > Cu_{Po} > Cu_{Py}$. The Ag content in sphalerite was 1 order of magnitude higher than that in pyrrhotite and was approximately 1 ppm in pyrite. The Ag distribution was $Ag_{Sph} > Ag_{Po} > Ag_{Py}$. Sphalerite was the main enrichment mineral of Sn, followed by pyrite. Sn was not uniformly distributed in sphalerite, pyrite, and pyrrhotite, and the Sn concentrations exhibited wide ranges in the three sulfides. The Sn distribution was $Sn_{Sph} > Sn_{Py} > Sn_{Po}$. The Pb contents in sphalerite, pyrite, and pyrrhotite also exhibited wide ranges and uneven distributions. The Pb distribution was $Pb_{Sph} > Pb_{Py} > Pb_{Po}$.

The Sb and W contents differed only slightly in sphalerite, pyrite, and pyrrhotite; some samples were below DL. The distributions of Sb and W were $Sb_{Py} > Sb_{Spy} > Sb_{Po}$ and $W_{Py} > W_{Po} > W_{Sph}$, respectively.

Thus, in the Huangshaping copper-polymetallic deposit, Ga, In, Cd, Mn, Cu, Ag, Sn, and Pb were enriched in sphalerite; Tl, Sb, W, and Se were relatively enriched in pyrite; and Ge was enriched in pyrrhotite. Sphalerite was the main carrier mineral of dispersed and rare elements.

### 5.2. Occurrence of trace elements in sulfides

LA-ICP-MS has a low detection limit that can accurately determine the concentration of elements in sulfides. Time-resolved LA-ICP-MS signal depth profile analysis was used to better explore the mode of trace element occurrence in minerals (Yuan et al., 2011; Hu et al., 2020; Quan et al., 2019; Wei et al., 2018).

As mentioned in Section 5.1, sphalerite, pyrite, and pyrrhotite were the carrier minerals for some of the dispersed and rare elements. Fe, Mn, and Cd were the main trace elements in sphalerite. They exhibited evident normal distributions on the trace element histogram (Fig. 8), indicating their uniform distributions within sphalerites. They exhibited smooth LA-ICP-MS time-resolved signal spectra (Fig. 9-a,b,c,d) and were consistent with the changes in Zn and S, indicating that these elements are similar to Zn and occur together in the sphalerite crystal lattice. The Ag and Cd signal curves in the LA-ICP-MS profiles were in good agreement, which may indicate that Ag and Cd enter the crystal lattice of sphalerite together. Sn and Cu did not exhibit normal distributions in the trace element histogram (Fig. 8). The signal curve changes of Cu and Sn were synergistic in the LA-ICP-MS profile. Moreover, in some samples, they appeared in the form of a smooth curve, which was consistent with the changes in other high-concentration elements (e.g., Fe, Mn, and Cd). However, the signal curves in the LA-ICP-MS profiles of some samples exhibited large simultaneous fluctuations, which were not caused by microscopic mineral inclusions (the mineral that enriched both Cu and Sn was stannite, which contains Fe; however, the Fe curve in the LA-ICP-MS profile was smooth, without fluctuations). Thus, these curves may indicate heterogeneous isomorphism of Cu and Sn. The Sn valence in sphalerite was not clear; however, $Sn^{2+}$, $Cu^{2+}$, and $Zn^{2+}$ had more similar ion radii and high geochemical affinity, and therefore, the most likely replacement would be $mCu^{2+}+nSn^{2+}\leftrightarrow(m+n)Zn^{2+}$. The trace element histogram and LA-ICP-MS profile of Ge show that it is isomorphic in sphalerite, and the Ge and Cu changes are similar but not identical. Ge exists in the form of $Ge^{4+}$ in sphalerite (Cook et al., 2015; Belissont et al., 2016). Based on previous studies (Hu et al., 2019; Wei et al., 2021), the most likely replacement would be $Ge^{4+}+2Cu^{+}\leftrightarrow3Zn^{2+}$. Although Pb concentration in sphalerite was very low, its curve change in the LA-ICP-MS profile was similar to that of Sb, and the Pb curve

**Table 1**
Trace elements in sulfide minerals from the Huangshaping deposit (ppm).

| Mineral | | S | Mn | Co | Ni | Cu | Ga | Ge | As | Se | Nb | Mo | Ag | Cd | In | Sn | Sb | W | Tl | Pb | Bi |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Level -96m | | | | | | | | | | | | | | | | | | | | | |
| HSP5 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 335452 | 1875 | 0.13 | 0.23 | 5269 | 10.9 | 2.92 | 0.34 | 2.72 | 0.0048 | 0.1 | 116 | 2207 | 1.32 | 20.8 | 0.83 | <DL | 0.079 | 41.3 | 0.02 |
| (n = 4) | Min | 334386 | 1712 | <DL | <DL | 1580 | 3.34 | 2.24 | <DL | <DL | <DL | <DL | 86.8 | 1908 | 0.99 | 6.36 | 0.13 | <DL | <DL | 10.6 | <DL |
| | Mean | 334823 | 1803 | 0.1 | 0.23 | 3601 | 8.34 | 2.64 | 0.34 | 2.72 | 0.0048 | 0.059 | 102 | 2010 | 1.1 | 13.6 | 0.42 | – | 0.057 | 19.6 | 0.02 |
| | S.D. | 449 | 71.5 | 0.037 | – | 1532 | 3.57 | 0.29 | – | – | – | 0.041 | 12.4 | 135 | 0.15 | 6.57 | 0.33 | – | 0.032 | 14.5 | – |
| Pyrrhotite | Max | 477366 | 0.8 | <DL | 0.92 | <DL | <DL | 11.3 | 0.76 | <DL | <DL | 1.12 | 4.52 | <DL | <DL | 0.17 | 0.56 | 0.2 | <DL | 4.52 | <DL |
| (n = 3) | Min | 444863 | <DL | <DL | 0.51 | <DL | <DL | 10.1 | <DL | <DL | <DL | <DL | 2.8 | <DL | <DL | <DL | 0.15 | <DL | <DL | 2.08 | <DL |
| | Mean | 462922 | 0.8 | – | 0.74 | – | – | 10.7 | 0.76 | – | – | 0.65 | 3.92 | – | – | 0.15 | 0.31 | 0.16 | – | 3.21 | – |
| | S.D. | 16550 | – | – | 0.21 | – | – | 0.62 | – | – | – | 0.66 | 0.97 | – | – | 0.04 | 0.21 | 0.058 | – | 1.23 | – |
| Level -136m | | | | | | | | | | | | | | | | | | | | | |
| HSP6 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 335795 | 2329 | 0.76 | 0.7 | 684 | 34.7 | 3.34 | 0.99 | 10.3 | 0.41 | 0.12 | 146 | 3350 | 115 | 4365 | 15.3 | 26.5 | 1.13 | 12403 | <DL |
| (n = 10) | Min | 331966 | 1131 | 0.12 | <DL | 65.2 | 2.16 | 2.3 | <DL | <DL | <DL | <DL | 48.6 | 2359 | 21 | 5.99 | 0.4 | <DL | 0.095 | 37.3 | <DL |
| | Mean | 333774 | 1725 | 0.42 | 0.43 | 240 | 17.5 | 2.74 | 0.88 | 10.3 | 0.22 | 0.11 | 96.1 | 2879 | 64 | 498 | 5.87 | 4.64 | 0.48 | 1476 | – |
| | S.D. | 1063 | 426 | 0.19 | 0.28 | 195 | 10.7 | 0.36 | 0.1 | – | 0.19 | 0.02 | 34.5 | 312 | 30.3 | 1290 | 4.98 | 9.78 | 0.34 | 3653 | – |
| Pyrite | Max | 534505 | 1472 | 1.11 | 5.07 | 2.88 | 0.07 | 9.96 | 23 | 4.76 | 0.0071 | 0.63 | 3.65 | 0.41 | 0.19 | 45.9 | 24.6 | 52.6 | 5.83 | 72.9 | 0.031 |
| (n = 6) | Min | 532765 | <DL | <DL | <DL | 0.49 | <DL | 8.32 | <DL | 1.09 | <DL | 0.06 | <DL | <DL | <DL | 1.21 | 1.14 | 5.81 | 2.4 | 4.87 | <DL |
| | Mean | 534197 | 371 | 1.01 | 2.69 | 1.49 | 0.045 | 9.04 | 8.21 | 2.93 | 0.0056 | 0.18 | 2.57 | 0.41 | 0.056 | 10.9 | 8.4 | 21.4 | 3.8 | 23.4 | 0.022 |
| | S.D. | 641 | 635 | 0.099 | 1.8 | 1.04 | 0.019 | 0.51 | 8.78 | 1.84 | 0.0015 | 0.21 | 0.95 | – | 0.069 | 15.7 | 8.41 | 16.1 | 1.11 | 24 | 0.0067 |
| HSP7 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 334377 | 1855 | 0.31 | 0.31 | 1645 | 21.8 | 3 | 0.59 | 15.7 | 0.025 | 0.11 | 116 | 3256 | 120 | 262 | 21.3 | 0.12 | 0.63 | 321 | 0.2 |
| (n = 7) | Min | 333036 | 794 | 0.12 | <DL | 596 | 16 | <DL | <DL | <DL | <DL | <DL | 45.8 | 2868 | 43 | 41.4 | 3.51 | <DL | <DL | 54.7 | <DL |
| | Mean | 333760 | 1207 | 0.21 | 0.31 | 1188 | 18.3 | 2.14 | 0.36 | 9.29 | 0.015 | 0.09 | 72.8 | 3040 | 84.7 | 114 | 11 | 0.11 | 0.35 | 156 | 0.12 |
| | S.D. | 441 | 334 | 0.07 | – | 328 | 1.93 | 0.39 | 0.14 | 4.51 | 0.0056 | 0.02 | 21.4 | 125 | 26.9 | 70 | 5.53 | 0.027 | 0.14 | 85.5 | 0.054 |
| Pyrite | Max | 534513 | 0.56 | 1.13 | 4.85 | 0.98 | 0.057 | 9.59 | 122 | 6.31 | 0.029 | 0.47 | 5.08 | <DL | 0.01 | 0.56 | 18.6 | 68.3 | 5.99 | 49.6 | 0.0062 |
| (n = 8) | Min | 534157 | <DL | <DL | <DL | <DL | <DL | 7.66 | <DL | <DL | <DL | <DL | 1.04 | <DL | <DL | <DL | <DL | <DL | <DL | 0.53 | <DL |
| | Mean | 534435 | 0.37 | 0.39 | 2.22 | 0.66 | 0.038 | 8.74 | 31 | 3.96 | 0.021 | 0.19 | 2.29 | – | 0.0083 | 0.29 | 6.99 | 18.1 | 3.23 | 9.4 | 0.0062 |
| | S.D. | 115 | 0.11 | 0.35 | 1.32 | 0.19 | 0.011 | 0.69 | 52.3 | 1.79 | 0.0083 | 0.13 | 1.41 | – | 0.0015 | 0.17 | 7.11 | 22.9 | 1.98 | 15.4 | – |
| HLR67 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 335333 | 1044 | 0.33 | 0.55 | 13324 | 58.5 | 3.34 | 0.42 | 5.2 | 0.013 | 0.047 | 83.8 | 2249 | 114 | 2769 | 1.46 | 0.36 | 0.46 | 80.2 | 28.8 |
| (n = 4) | Min | 334630 | 452 | 0.26 | <DL | 2125 | 5.24 | 2.37 | <DL | 2.59 | <DL | <DL | 18.1 | 2015 | 53.3 | 7.59 | <DL | 0.027 | <DL | 2.94 | 0.64 |
| | Mean | 335001 | 797 | 0.29 | 0.35 | 6500 | 29 | 2.74 | 0.37 | 4.13 | 0.013 | 0.047 | 38.9 | 2143 | 86.6 | 786 | 0.76 | 0.12 | 0.2 | 22.4 | 8.12 |
| | S.D. | 250 | 215 | 0.03 | 0.2 | 4149 | 21.9 | 0.36 | 0.051 | 0.95 | – | – | 26.3 | 91.4 | 23.9 | 1152 | 0.69 | 0.14 | 0.19 | 33.4 | 12 |
| Pyrrhotite | Max | 534515 | 1475 | 0.3 | 2.39 | 7902 | 6.27 | 12.4 | 0.4 | 9.52 | 0.014 | 0.31 | 60.5 | 866 | 59.9 | 247 | 2.5 | 2.64 | 1.06 | 151 | 52.6 |
| (n = 13) | Min | 401631 | <DL | 0.14 | 0.98 | <DL | <DL | 8.17 | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | 0.41 | 0.07 |
| | Mean | 449729 | 247 | 0.21 | 1.78 | 1581 | 1.62 | 10.6 | 0.4 | 4.93 | 0.011 | 0.19 | 8.72 | 434 | 10 | 27.9 | 0.6 | 0.97 | 0.46 | 14 | 5.39 |
| | S.D. | 30257 | 549 | 0.046 | 0.36 | 3160 | 2.69 | 1.03 | – | 2.8 | 0.0021 | 0.089 | 18.3 | 431 | 22.3 | 77.4 | 0.95 | 1.04 | 0.42 | 39.5 | 13.8 |
| Level -156m | | | | | | | | | | | | | | | | | | | | | |
| HSP8 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 334892 | 1400 | 0.89 | 0.65 | 4518 | 11.9 | 2.86 | 0.82 | 5.76 | 0.35 | 0.12 | 105 | 3206 | 157 | 1417 | 13.8 | 0.23 | 0.52 | 6669 | 0.32 |
| (n = 5) | Min | 331879 | 1267 | 0.67 | <DL | 182 | 5.74 | 2.13 | <DL | <DL | <DL | <DL | 16.7 | 2917 | 36.7 | 5.09 | <DL | <DL | 0.037 | 2.46 | <DL |
| | Mean | 333698 | 1362 | 0.79 | 0.46 | 1862 | 7.85 | 2.55 | 0.44 | 5.62 | 0.19 | 0.085 | 54.2 | 3108 | 87.4 | 787 | 6.92 | 0.14 | 0.24 | 1436 | 0.31 |
| | S.D. | 1025 | 48.7 | 0.071 | 0.16 | 1525 | 2.16 | 0.25 | 0.29 | 0.14 | 0.16 | 0.024 | 33.1 | 104 | 38.9 | 641 | 5.41 | 0.076 | 0.19 | 2622 | 0.0097 |
| Pyrite | Max | 534502 | 0.38 | 1.26 | 4.19 | 1.28 | 0.086 | 9.12 | <DL | 20.9 | 0 | 0.17 | 2.73 | 0.12 | 0.0052 | 1.05 | 0.66 | 32.1 | 31.2 | 23.4 | 0.012 |
| (n = 4) | Min | 534471 | <DL | <DL | <DL | <DL | 0.019 | 8.49 | <DL | 1.96 | <DL | 0.038 | 0.7 | <DL | <DL | 0.16 | 0.12 | 5.51 | 12.5 | 4 | <DL |
| | Mean | 534486 | 0.37 | 0.98 | 2.88 | 0.83 | 0.049 | 8.8 | – | 12.4 | 0 | 0.095 | 1.6 | 0.12 | 0.0046 | 0.59 | 0.43 | 20.7 | 20.7 | 12.6 | 0.012 |
| | S.D. | 11.5 | 0.077 | 0.25 | 0.95 | 0.33 | 0.029 | 0.22 | – | 7.3 | 0 | 0.054 | 0.73 | – | 0.0005 | 0.39 | 0.2 | 10.8 | 7.19 | 7.07 | – |

(*continued on next page*)

**Table 1** (*continued*)

| Mineral | | S | Mn | Co | Ni | Cu | Ga | Ge | As | Se | Nb | Mo | Ag | Cd | In | Sn | Sb | W | Tl | Pb | Bi |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pyrrhotite | Max | 789176 | 1.3 | <DL | <DL | 4.16 | 0.041 | 11.7 | <DL | 46.4 | 0.016 | 0.17 | 11.5 | 0.29 | 0.023 | 3.25 | 5.2 | <DL | 1.11 | 173 | 0.081 |
| (n = 3) | Min | 490747 | <DL | <DL | <DL | 4.05 | <DL | 10.7 | <DL | 22.1 | <DL | <DL | 2.46 | <DL | 0.007 | <DL | <DL | <DL | <DL | 1.83 | <DL |
| | Mean | 658044 | 1.13 | – | – | 4.11 | 0.041 | 11.3 | – | 33.4 | 0.016 | 0.12 | 8.21 | 0.29 | 0.015 | 2.87 | 3.59 | – | 0.91 | 91 | 0.059 |
| | S.D. | 124488 | 0.16 | – | – | 0.055 | – | 0.42 | – | 10 | – | 0.05 | 4.07 | – | 0.0064 | 0.38 | 1.61 | – | 0.2 | 69.9 | 0.021 |
| HSP9 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 334766 | 1759 | 0.62 | 0.35 | 1116 | 21.7 | 3.45 | 0.58 | 18.4 | 0.019 | 0.22 | 27.1 | 3728 | 187 | 888 | 5.26 | 0.079 | 0.61 | 62.4 | 0.094 |
| (n = 21) | Min | 332607 | 1009 | 0.32 | <DL | 54.7 | 3.81 | 1.4 | <DL | <DL | <DL | <DL | 7.69 | 2499 | 18.8 | 0.39 | <DL | <DL | <DL | 1.61 | <DL |
| | Mean | 334104 | 1381 | 0.46 | 0.21 | 250 | 9.11 | 2.57 | 0.26 | 6.93 | 0.0088 | 0.09 | 15.5 | 3183 | 82.8 | 69 | 1.31 | 0.047 | 0.17 | 11.5 | 0.054 |
| | S.D. | 507 | 187 | 0.089 | 0.1 | 269 | 4.42 | 0.55 | 0.15 | 4.59 | 0.0057 | 0.05 | 5.25 | 267 | 42.6 | 189 | 1.39 | 0.021 | 0.15 | 13 | 0.031 |
| Pyrrhotite | Max | 607357 | <DL | 0.75 | 21.9 | <DL | 0.051 | 13.7 | 1.44 | 9.64 | 0.025 | 0.14 | 8.78 | 0.21 | 0.054 | 3.51 | 7.05 | 21.7 | 3.58 | 241 | 0.79 |
| (n = 10) | Min | 444047 | <DL | <DL | <DL | <DL | <DL | 9.56 | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | 0.11 | <DL |
| | Mean | 489156 | - | 0.38 | 6.2 | - | 0.051 | 11.2 | 1.12 | 9.64 | 0.023 | 0.14 | 3.41 | 0.21 | 0.054 | 1.83 | 4.13 | 11.9 | 1.09 | 30.3 | 0.42 |
| | S.D. | 46660 | - | 0.22 | 5.68 | - | - | 1.2 | 0.29 | - | 0.0013 | - | 2.84 | - | - | 1.68 | 2.65 | 9.78 | 1.45 | 71.4 | 0.3 |
| Level -176m | | | | | | | | | | | | | | | | | | | | | |
| HSP10 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 334968 | 1695 | 0.8 | 0.46 | 2158 | 16 | 2.93 | 0.37 | 1.64 | 0.098 | 0.085 | 59.6 | 2908 | 67.8 | 574 | 6.26 | 0.068 | 0.28 | 150 | 0.044 |
| (n = 8) | Min | 333922 | 1352 | 0.56 | <DL | 87.4 | 8.89 | 2.24 | <DL | <DL | <DL | <DL | 11.4 | 2464 | 7.42 | 0.86 | <DL | <DL | <DL | 1.81 | <DL |
| | Mean | 334565 | 1569 | 0.64 | 0.25 | 751 | 12.6 | 2.62 | 0.24 | 1.43 | 0.0088 | 0.044 | 30.3 | 2588 | 23.4 | 81.1 | 2.44 | 0.042 | 0.2 | 43.1 | 0.031 |
| | S.D. | 393 | 125 | 0.08 | 0.13 | 756 | 2.91 | 0.2 | 0.073 | 0.15 | 0.001 | 0.021 | 15.5 | 141 | 19.5 | 187 | 2.03 | 0.019 | 0.078 | 46.5 | 0.0095 |
| Pyrrhotite | Max | 565595 | 8.51 | 0.19 | 5.4 | <DL | 0.089 | 14.4 | 0.51 | <DL | 0.014 | 0.3 | 11.6 | 0.36 | 0.015 | 0.1 | 3.57 | <DL | 0.24 | 26.5 | <DL |
| (n = 13) | Min | 446683 | <DL | <DL | <DL | <DL | <DL | 10.4 | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL |
| | Mean | 483825 | 8.51 | 0.16 | 2.5 | – | 0.049 | 11.3 | 0.51 | – | 0.014 | 0.16 | 3.81 | 0.32 | 0.015 | 0.094 | 2.09 | – | 0.15 | 5.11 | – |
| | S.D. | 36128 | – | 0.027 | 1.26 | – | 0.019 | 0.98 | – | – | – | 0.064 | 3.52 | 0.052 | 0.0026 | 0.0059 | 1.48 | – | 0.064 | 6.78 | – |
| HSP11 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 335157 | 1739 | 0.39 | 0.24 | 1645 | 30.6 | 4.01 | 4.12 | 2.47 | 0.21 | 0.11 | 322 | 2811 | 7 | 123 | 27.6 | 0.85 | 3.62 | 1685 | 0.024 |
| (n = 6) | Min | 329638 | 370 | 0.23 | <DL | 112 | 2.98 | 2.18 | <DL | <DL | <DL | <DL | 58.6 | 2435 | 0.19 | 6.13 | 0.7 | <DL | 0.15 | 83.8 | <DL |
| | Mean | 333261 | 1216 | 0.31 | 0.23 | 951 | 10.3 | 2.89 | 1.72 | 2.47 | 0.059 | 0.076 | 142 | 2592 | 2.74 | 33.6 | 6.47 | 0.29 | 1.19 | 457 | 0.024 |
| | S.D. | 2069 | 471 | 0.059 | 0.011 | 529 | 9.6 | 0.7 | 1.74 | – | 0.086 | 0.03 | 92 | 119 | 2.35 | 41.1 | 9.63 | 0.33 | 1.22 | 562 | – |
| Pyrite | Max | 534430 | 812 | 0.48 | 6.47 | 4.78 | 0.14 | 9.6 | 44.3 | <DL | 0.0079 | 0.19 | 9.01 | <DL | 0.03 | 4.85 | 26.7 | 11.4 | 23.2 | 93.9 | 0.031 |
| (n = 4) | Min | 528543 | 2.66 | <DL | <DL | 2.22 | 0.03 | 7.83 | 0.39 | <DL | <DL | <DL | 1.49 | <DL | <DL | 1.88 | 1.81 | 0.82 | 2.74 | 17.3 | <DL |
| | Mean | 532273 | 308 | 0.3 | 3.95 | 3.15 | 0.085 | 8.64 | 22.7 | – | 0.0079 | 0.11 | 4.12 | – | 0.024 | 3.46 | 17 | 3.63 | 9.32 | 54.6 | 0.017 |
| | S.D. | 2290 | 318 | 0.2 | 2.74 | 1.16 | 0.039 | 0.64 | 21.5 | – | – | 0.063 | 2.94 | – | 0.0051 | 1.14 | 9.97 | 4.51 | 8.32 | 30.5 | 0.013 |
| Level -216m | | | | | | | | | | | | | | | | | | | | | |
| HSP12 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 335331 | 1407 | 0.56 | 0.37 | 1928 | 10 | 3.1 | <DL | 2.67 | 0.0086 | 0.062 | 67.6 | 2981 | 60.5 | 70.8 | 5.02 | 0.039 | 0.58 | 148 | 17.5 |
| (n = 7) | Min | 334724 | 974 | 0.41 | <DL | 99 | 2.49 | 1.61 | <DL | <DL | <DL | <DL | 15.6 | 2508 | 12.9 | 1.64 | 0.24 | <DL | 0.015 | 7.48 | 0.13 |
| | Mean | 335016 | 1229 | 0.48 | 0.22 | 613 | 6.35 | 2.54 | – | 2.4 | 0.0061 | 0.045 | 27.4 | 2794 | 29.5 | 19.5 | 2.05 | 0.039 | 0.14 | 37.5 | 4.45 |
| | S.D. | 206 | 132 | 0.043 | 0.1 | 717 | 2.27 | 0.45 | – | 0.28 | 0.0018 | 0.017 | 16.8 | 139 | 14.5 | 22.5 | 1.38 | – | 0.18 | 46.1 | 5.74 |
| Pyrite | Max | 534506 | 205 | 1.52 | 40 | 1.05 | 0.14 | 8.57 | 0.32 | 1.1 | 0.019 | 0.51 | 2.24 | <DL | 0.35 | 70.5 | 4.49 | 47.6 | 1.96 | 21.8 | 5.61 |
| (n = 6) | Min | 532686 | 0.33 | 0.35 | 8.88 | <DL | <DL | 8.06 | <DL | <DL | <DL | 0.03 | <DL | <DL | <DL | 0.12 | 0.27 | 17.9 | 0.28 | 2.17 | <DL |
| | Mean | 534072 | 41.4 | 0.86 | 22.6 | 0.86 | 0.079 | 8.33 | 0.21 | 1.1 | 0.0097 | 0.16 | 1.2 | – | 0.35 | 14.4 | 1.55 | 30.9 | 1.07 | 9.4 | 1.47 |
| | S.D. | 702 | 81.9 | 0.38 | 10 | 0.16 | 0.063 | 0.22 | 0.09 | – | 0.0057 | 0.18 | 0.64 | – | – | 28 | 1.56 | 10.9 | 0.66 | 7.12 | 2.39 |
| HSP13 | | | | | | | | | | | | | | | | | | | | | |
| Sphalerite | Max | 334877 | 700 | 0.13 | 0.16 | 35121 | 30.7 | 2.89 | 0.31 | 1.69 | 0.0105 | 0.025 | 135 | 3966 | 425 | 603 | 1.11 | 0.058 | 0.49 | 182 | 1.04 |
| (n = 6) | Min | 333139 | 519 | 0.067 | <DL | 1943 | 2.32 | 1.28 | <DL | <DL | <DL | <DL | 49.1 | 3552 | 12.9 | 27.4 | 0.007 | <DL | <DL | 2.94 | 0.0074 |
| | Mean | 334105 | 641 | 0.093 | 0.098 | 14017 | 9.37 | 2.09 | 0.16 | 1.07 | 0.0024 | 0.025 | 88.5 | 3748 | 142 | 238 | 0.29 | 0.013 | 0.13 | 57.9 | 0.49 |
| | S.D. | 580 | 60.8 | 0.018 | 0.049 | 11070 | 10.1 | 0.56 | 0.1 | 0.42 | 0.0039 | – | 30.8 | 151 | 172 | 192 | 0.38 | 0.02 | 0.18 | 59.6 | 0.46 |
| Pyrite | Max | 534511 | 0.62 | 0.82 | 10.5 | 3.91 | 0.038 | 10.3 | 1.62 | 9.53 | 0.011 | 0.16 | 3.2 | 0.13 | 0.01 | 0.93 | 6.33 | 54.4 | 1.59 | 38.8 | 1.82 |
| (n = 10) | Min | 534131 | <DL | 0.061 | 0.88 | <DL | <DL | 7.68 | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | <DL | 17.7 | 0.09 | 0.8 | <DL |

(*continued on next page*)

**Table 1** (*continued*)

| Mineral | | S | Mn | Co | Ni | Cu | Ga | Ge | As | Se | Nb | Mo | Ag | Cd | In | Sn | Sb | W | Tl | Pb | Bi |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | 534453 | 0.45 | 0.4 | 5.38 | 1.48 | 0.029 | 8.48 | 0.7 | 4.95 | 0.0051 | 0.074 | 1.68 | 0.13 | 0.0083 | 0.37 | 2.24 | 33.1 | 0.77 | 12.9 | 0.55 |
| | S.D. | 110 | 0.11 | 0.22 | 3.03 | 1.04 | 0.0054 | 0.71 | 0.38 | 2.52 | 0.0034 | 0.044 | 0.73 | – | 0.0019 | 0.3 | 1.92 | 12.9 | 0.51 | 10.3 | 0.7 |

Postscript: <DL is below the detection limit.

primarily exhibited a single peak, indicating that it may be caused by the late stage mineral assemblage (galena and minerals which enrich Sn and Sb) (Fig. 5). The trace element histogram of In exhibited an evident normal distribution. In the LA-ICP-MS profile, the signal curve of In is smooth, which is consistent with the changes in Zn, S, Cd, and Cu. Although the Ga concentration in sphalerite was low, its trace element histogram exhibited a normal distribution. Additionally, the Ga, In, and Cu signal curves exhibit good synergisms in the LA-ICP-MS profile. The chemical properties of $In^{3+}$ and $Ga^{3+}$ were similar to those of $Zn^{2+}$, and the $Zn^{2+}$ in sphalerite was easily replaced by the similar ion: $In^{3+}(Ga^{3+}) + Cu^{+} \leftrightarrow 2Zn^{2+}$. However, some sphalerite samples had high In and low Cu concentrations, whereas the opposite is true in other samples, indicating a negative correlation between the two concentrations. The reason for this phenomenon may be that $Cu^{+}$ can replace $Zn^{2+}$ with many paired ions, and $Cu^{+}$ is more likely to combine with other ions than with $In^{3+}$. The elemental properties of In and Sn are similar, and they are characterized by synchronous migration and enrichment in hydrothermal fluid (Li et al., 2019d; Wang et al., 2019; Li et al., 2020b). This phenomenon may also be caused by the heterogeneous substitution of Sn.

There were a few trace elements in pyrite. The signal curves of Tl, Sb, Ag, and Pb in pyrite show consistent changes in their LA-ICP-MS profiles (Fig. 9-e,f), which may be caused by late stage galena replacing the early pyrite. W existed in a special form in pyrite, and the trace element histogram suggests that it is not uniformly distributed (Fig. 10). The LA-ICP-MS profile presents a relatively smooth signal curve or no signal at all. Based on the pyrite genesis, this could be related to W from the residual magmatic hydrothermal fluid that experienced W–Mo(Fe) mineralization and that occurred in pyrite as a solid solution.

Owing to the defects in the pyrrhotite crystal structure, other elements or mineral inclusions can easily enter the crystal lattice. Therefore, with increased laser erosion depth, other mineral inclusions are easily denuded. Moreover, as pyrrhotite is an early forming mineral, it is replaced and altered by later minerals. Therefore, the LA-ICP-MS profile of pyrrhotite is relatively complex (Fig. 9-g,h). Considering these reasons, only those areas with stable and reasonable LA-ICP-MS profile curves were selected for LA-ICP-MS analysis of pyrrhotite,. Except for Ge, most other elements in pyrrhotite were below DL, the ranges of Ge concentrations were small, and the distribution was relatively uniform (Fig. 10). Combined with the analysis of the LA-ICP-MS profile, this indicates that Ge may incorporate into pyrrhotite through coupling substitution. The Ag signal curve appears as a peak value in the LA-ICP-MS profile of some samples, which may be caused by laser erosion into the mineral micro-inclusions of Ag. The Se distribution was uneven (Fig. 10). According to the LA-ICP-MS profile, the peaks of the Se, Pb, and Ag signal curves are more consistent. When these peaks appear, the Fe signal curve shows a valley. The Se anomaly in pyrite was similar to that in pyrrhotite, which may be caused by late stage galena replacing pyrrhotite.

To summarize, in sphalerite, Fe, Mn, Cd, Ga, In, Cu, Sn, and Ag exist by substituting for other elements, whereas Pb and Sb are caused by late stage mineral assemblage (e.g., galena) replacement. In pyrite, W occurs as a solid solution; Pb is caused by late stage galena replacing; and Tl, Sb, and Ag occur in the late stage galena. In pyrrhotite, Ge is incorporated through coupling substitution, Pb is caused by late-stage galena replacing, and Ag and Se occur in late-stage galena.

### 5.3. Ore-forming temperature

Previous studies have ascertained the ore-forming temperature via many different methods, including fluid inclusion thermometry, geothermometer methods (ZnS–FeS geothermometry (Barton and Toulmin, 1966; Scott and Barnes, 1971; Lu, 1975a), and sphalerite–galena sulfur isotope geothermometry (Grootenboer and Schwarcz, 1969; Rye, 1974; lu, 1975b; Ding et al., 1992). Many studies have demonstrated that some trace elements contained in sphalerite are sensitive to temperature

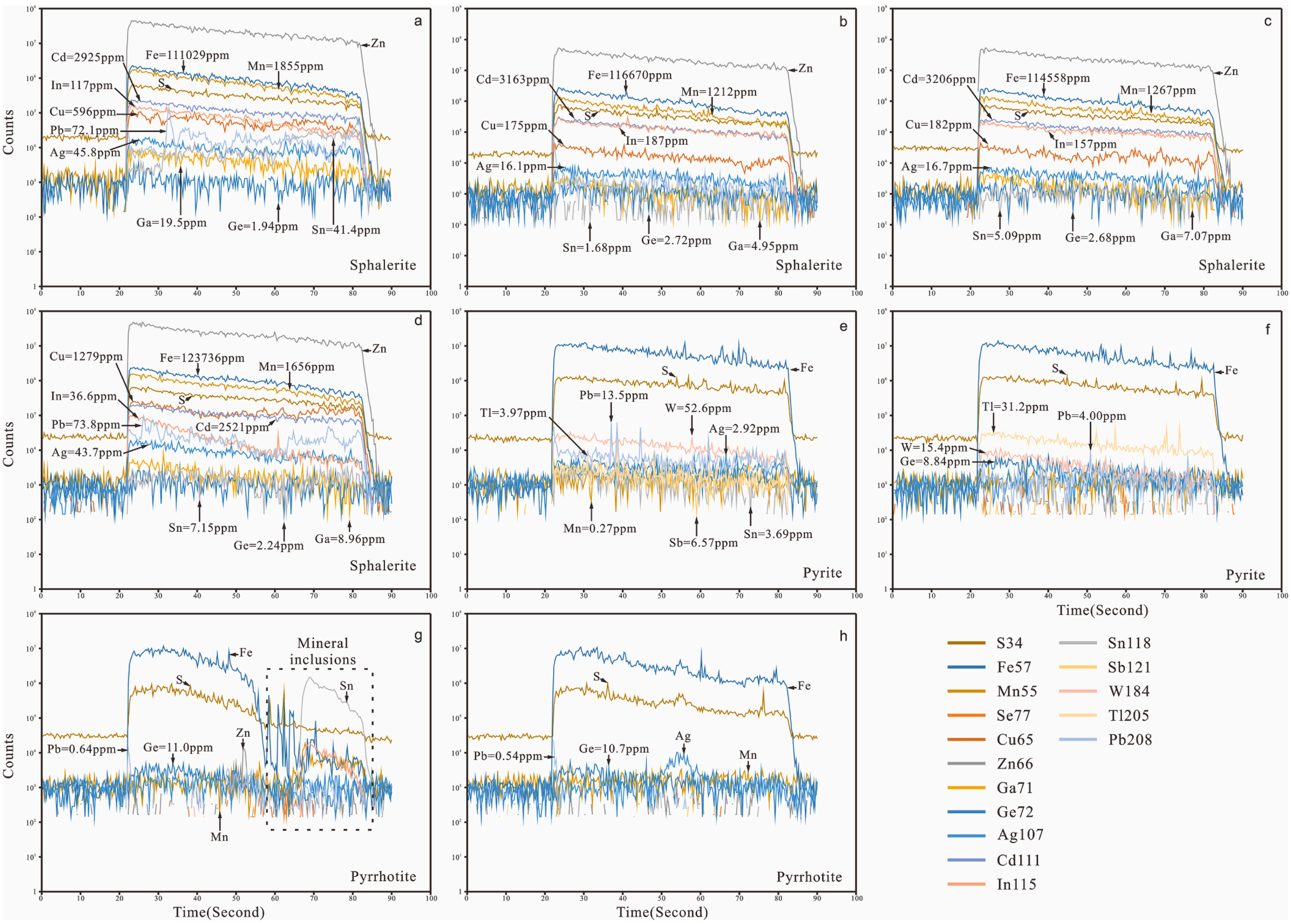


**Fig. 9.** Time-resolved LA-ICP-MS signal depth profiles. a - HSP7Sph1; b - HSP9Sph210; c - HSP8Sph3; d - HSP10Sph7; e - HSP6Py2; f - HSP8Py3; g - HLR67Po2; h - HSP10Po4.

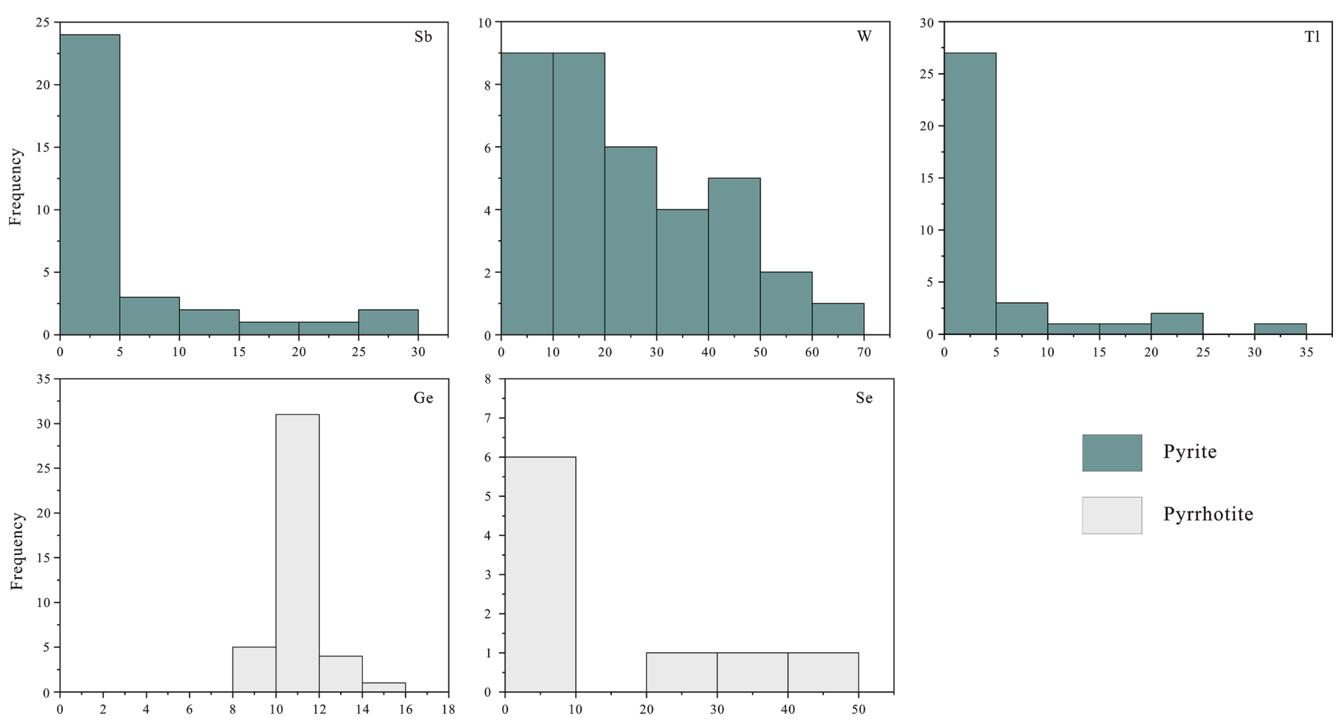


**Fig. 10.** Frequency histogram of the contents of pyrite and pyrrhotite elements.

**Table 2**
Calculation of sphalerite formation temperature.

| Sample | Temperature | Max | Min |
|---|---|---|---|
| Level −96 m | | | |
| HSP-5-sp − 1 | 313 ± 24℃ | 313 ± 24℃ | 300 ± 22℃ |
| HSP-5-sp − 2 | 300 ± 22℃ | | |
| HSP-5-sp − 3 | 307 ± 23℃ | | |
| HSP-5-sp − 4 | 302 ± 23℃ | | |
| Level −136 m | | | |
| HSP-6-sp-1–1 | 313 ± 24℃ | 326 ± 26℃ | 294 ± 22℃ |
| HSP-6-sp-1–2 | 316 ± 24℃ | | |
| HSP-6-sp-1–3 | 313 ± 24℃ | | |
| HSP-6-sp-1–5 | 315 ± 24℃ | | |
| HSP-6-sp-1–6 | 308 ± 23℃ | | |
| HSP-6-sp-1–7 | 326 ± 26℃ | | |
| HSP-6-sp-1–8 | 316 ± 25℃ | | |
| HSP-6-sp-1–9 | 315 ± 24℃ | | |
| HSP-6-sp-1–10 | 322 ± 25℃ | | |
| HSP-6-sp-1–11 | 323 ± 25℃ | | |
| HSP-7-sp-1–1 | 318 ± 25℃ | | |
| HSP-7-sp-1–2 | 308 ± 23℃ | | |
| HSP-7-sp-1–3 | 315 ± 24℃ | | |
| HSP-7-sp-1–5 | 315 ± 24℃ | | |
| HSP-7-sp-1–6 | 314 ± 24℃ | | |
| HSP-7-sp-1–7 | 325 ± 26℃ | | |
| HLR-67-sp-1–2 | 298 ± 22℃ | | |
| HLR-67-sp-1–3 | 297 ± 22℃ | | |
| HLR-67-sp-1–4 | 317 ± 25℃ | | |
| HLR-67-sp-1–6 | 294 ± 22℃ | | |
| Level −156 m | | | |
| HSP-8-sp − 1 | 323 ± 25℃ | 341 ± 28℃ | 308 ± 23℃ |
| HSP-8-sp − 2 | 330 ± 26℃ | | |
| HSP-8-sp − 3 | 326 ± 26℃ | | |
| HSP-8-sp − 4 | 329 ± 26℃ | | |
| HSP-8-sp − 5 | 325 ± 26℃ | | |
| HSP-9-sp1 − 1 | 333 ± 27℃ | | |
| HSP-9-sp1 − 2 | 329 ± 26℃ | | |
| HSP-9-sp1 − 4 | 337 ± 27℃ | | |
| HSP-9-sp1 − 5 | 333 ± 27℃ | | |
| HSP-9-sp2 − 1 | 327 ± 26℃ | | |
| HSP-9-sp2 − 2 | 321 ± 25℃ | | |
| HSP-9-sp2 − 3 | 328 ± 26℃ | | |
| HSP-9-sp1 − 6 | 308 ± 23℃ | | |
| HSP-9-sp1 − 7 | 324 ± 26℃ | | |
| HSP-9-sp2 − 4 | 327 ± 26℃ | | |
| HSP-9-sp2 − 5 | 320 ± 25℃ | | |
| HSP-9-sp2 − 6 | 310 ± 24℃ | | |
| HSP-9-sp2 − 8 | 324 ± 26℃ | | |
| HSP-9-sp1 − 10 | 329 ± 26℃ | | |
| HSP-9-sp1 − 11 | 341 ± 28℃ | | |
| HSP-9-sp1 − 12 | 316 ± 24℃ | | |
| HSP-9-sp2 − 9 | 313 ± 24℃ | | |
| HSP-9-sp2 − 10 | 313 ± 24℃ | | |
| HSP-9-sp1 − 8 | 333 ± 27℃ | | |
| HSP-9-sp2 − 7 | 335 ± 27℃ | | |
| HSP-9-sp1 − 9 | 321 ± 25℃ | | |
| Level −176 m | | | |
| HSP-10-sp − 1 | 306 ± 23℃ | 325 ± 26℃ | 288 ± 21℃ |
| HSP-10-sp − 2 | 315 ± 24℃ | | |
| HSP-10-sp − 3 | 313 ± 24℃ | | |
| HSP-10-sp − 4 | 306 ± 23℃ | | |
| HSP-10-sp − 5 | 311 ± 24℃ | | |
| HSP-10-sp − 6 | 325 ± 26℃ | | |
| HSP-10-sp − 7 | 315 ± 24℃ | | |
| HSP-10-sp − 8 | 299 ± 22℃ | | |
| HSP-11-sp − 1 | 308 ± 23℃ | | |
| HSP-11-sp − 2 | 307 ± 23℃ | | |
| HSP-11-sp − 3 | 288 ± 21℃ | | |
| HSP-11-sp − 4 | 304 ± 23℃ | | |
| HSP-11-sp − 5 | 302 ± 23℃ | | |
| HSP-11-sp − 6 | 325 ± 26℃ | | |
| Level −216 m | | | |
| HSP-12-sp − 1 | 319 ± 25℃ | 329 ± 26℃ | 313 ± 24℃ |
| HSP-12-sp − 2 | 323 ± 25℃ | | |
| HSP-12-sp − 3 | 329 ± 26℃ | | |
| HSP-12-sp − 4 | 324 ± 26℃ | | |
| HSP-12-sp − 5 | 324 ± 26℃ | | |
| HSP-12-sp − 6 | 325 ± 26℃ | | |
| HSP-12-sp − 7 | 324 ± 26℃ | | |
| HSP-13-sp − 1–1 | 326 ± 26℃ | | |
| HSP-13-sp − 2–1 | 318 ± 25℃ | | |
| HSP-13-sp − 3–1 | 325 ± 26℃ | | |
| HSP-13-sp − 4–1 | 319 ± 25℃ | | |
| HSP-13-sp − 5–1 | 313 ± 24℃ | | |
| HSP-13-sp − 6–1 | 320 ± 25℃ | | |

(Cook et al., 2009; Ye et al., 2011; Liu et al., 2012; Ye et al., 2012), and hence, these trace elements can be used to calculate the ore-forming temperature.

Sphalerite obtained from medium–high-temperature deposits was characterized by a high concentration of high-temperature ore-forming elements (e.g., Fe, Mn, and In) and high In/Ga ratios (Zhang, 1987; Ye et al., 2012; Leng and Qi, 2017). Sphalerites formed by low-temperature deposits are characterized by the enrichment of low-temperature ore-forming elements (e.g., Cd, Ge, and Ga) and low In/Ge ratios (Ye et al., 2016; Hu et al., 2019; Guo et al., 2020). Therefore, the ore-forming temperature of a deposit can be estimated based on the Fe and Mn contents and element ratios (In/Ge and In/Ga) of sphalerite. Sphalerite from the Huangshaping deposit was characterized by Fe and Mn enrichment, with the Fe content ranging 87,523–146,111 ppm. The Mn content ranged from 370 to 2,329 ppm, which is similar to that of a medium-temperature deposit (low-temperature genesis of Tianbaoshan Pb–Zn deposit in Sichuan, Mn: 1.92–20.1 ppm, Ye et al., 2016; low-temperature genesis of Maliping Pb–Zn deposit in northeast Yunnan, Mn: 4.32–881 ppm, Hu et al., 2019; medium–high-temperature genesis of Laochang Pb–Zn polymetallic deposit in Yunnan, Mn: 1715–4,152 ppm, Ye et al., 2012; medium-temperature genesis of Hetaoping Pb–Zn polymetallic deposit in Yunnan, Mn: 1254–5,485 ppm, Ye et al., 2011). The Zn/Cd ratio ranged from 135 to 275, with an average of 190 ($n = 78$), which is similar to the Zn/Cd ratio ($100 < Zn/Cd < 500$) of sphalerite formed at medium–high temperature (Song, 1982). The ratios of In/Ge (0.082–203, averaging 26.2, $n = 77$) and In/Ga (0.060–37.7, averaging 7.14, $n = 78$) were high. These ratios are similar to those of the Laochang Pb–Zn polymetallic deposit (In/Ge: 11–1,689, In/Ga: 0.88–99.6) in Yunnan but significantly lower than those of the Goutouling deposit (In/Ge: 2,091–16,923, In/Ga: 149.8–792.7) in the Furong orefield.

Frenzel et al. (2016) proposed a new geothermometer based on the relationship between Ga, Ge, In, Fe, and Mn contents of sphalerite, which was initially used to determine the ore-forming temperature (Bauer et al., 2018; Hu et al., 2020; Knorsch et al., 2020; Wei et al., 2021). According to this geothermometer calculation, the sphalerite formation temperature in the Huangshaping deposit ranged from 288 °C to 341 °C (Table 2). Additionally, sphalerite in the Huangshaping deposit enriched medium–high-temperature ore-forming elements. Thus, the sphalerite in the Huangshaping deposit was precipitated in medium–high-temperature conditions. This is consistent with previous studies about fluid inclusions (homogeneous temperature of the fluid inclusions was associated with Cu–Mo–Zn–Pb mineralization ranging from 160 °C to 350 °C, Huang et al., 2013; Li et al., 2019c). There was no significant difference in the formation temperature of sphalerite at different levels in the deposit (Table 2), which may be due to two factors: (1) the error range of the thermometer is large and the formation temperature cannot be accurately defined, or (2) the sphalerite was precipitated during a rapid hydrothermal event.

## 5.4. Ore genesis

It is widely accepted that the Co/Ni and S/Se ratios of pyrite can be used to constrain its genesis (Liu et al., 2019; Li et al., 2020a; Huang et al., 2021; Liu et al., 2021). Ni enters silicate minerals more easily than

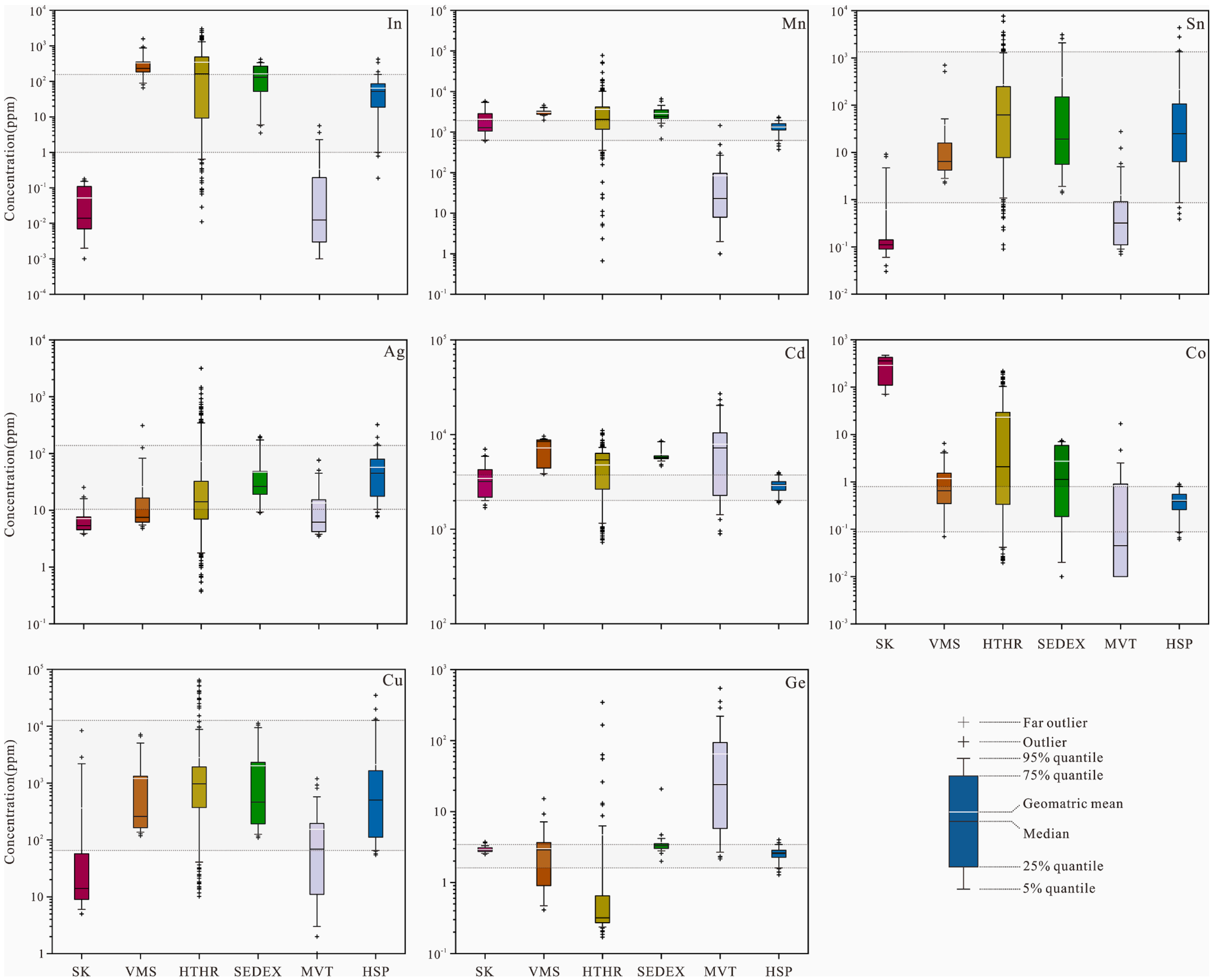


**Fig. 11.** In-situ minor/trace elements in sphalerite from the Huangshaping deposit (HSP) compared with high-temperature hydrothermal (HTHR), skarn (SK), VMS, SEDEX, and MVT deposits. Data from Bauer et al. (2018); Cook et al. (2009); Hu et al. (2020); Ye et al. (2011).

Co, and Co is more enriched in magmatic hydrothermal fluids; therefore, the Co/Ni ratio of magmatic pyrite is generally greater than 1.00, whereas the Co/Ni ratio of skarn pyrite varies widely but is generally close to 1 (Bralia et al., 1979; Leng, 2017; Liu et al., 2019). The S/Se ratio is mostly in the range of 260,000–500,000 in sedimentary pyrite but is lower in hydrothermal pyrite (Xu and Shao, 1980). In this study, the Co/Ni ratio (0.036–0.41, mean = 0.13, n = 29) of pyrite that formed during the Pb–Zn mineralization stage was lower than 1 but between 1 and 0.1, which is higher than that of sedimentary pyrite. This may indicate that the pyrite was crystallized during the early stage, resulting in a low Co/Ni ratio of the residual hydrothermal fluid. The S/Se ratio of pyrite exhibited a wide range (25,581–488,380, averaging 173905, n = 19) but is generally lower than that of sedimentary pyrite. Additionally, pyrite was deficient in low-temperature elements (e.g., As, Te, and Ag) and rich in medium–high-temperature elements (e.g., Mn and W). The significance of the pyrrhotite Co/Ni ratio has not yet been determined; however, several studies have used it (Leng, 2017; Liu et al., 2019). The Co/Ni ratio of pyrrhotite that formed during the Pb–Zn mineralization stage ranged from 0.035 to 0.20 (averaging 0.10, n = 26), which was lower than that of pyrite. The Co/Ni ratio of pyrrhotite was less than that of pyrite because Ni entered pyrite when it crystallized during stage V. Compared with pyrite, pyrrhotite was rich in medium-temperature ore-forming elements (e.g., Zn, Pb, and Ge), which may be the result of continuous cooling of the hydrothermal fluid. This evidence suggests that pyrite and pyrrhotite were not directly formed by magmatic hydrothermal fluid or skarn mineralization but may have formed by residual magmatic hydrothermal fluid, which is related to skarn mineralization.

The sphalerite formation temperature is relatively high, and the elements contained in sphalerite are not easily altered. Therefore, the trace elements of sphalerite can accurately constrain the genesis of the deposit (Ye et al., 2011; Belissont et al., 2014; Leng and Qi, 2017; Wei et al., 2021). The trace elements of sphalerite in the Huangshaping deposit were compared with those of many other types of deposits (Skarn, Sedex, VMS, MVT, and high-temperature hydrothermal deposits) (Fig. 11). Sphalerite from the Huangshaping deposit exhibited similar trace element ranges to that of the high-temperature hydrothermal deposits and skarn deposits. It was enriched in Fe, Ag, Mn, Sn, In, and Cu but was depleted in Cd, Ge, and Co, unlike the sphalerite composition in MVT deposits. Ye et al. (2011) proposed several binary diagrams for discriminating the genesis of deposits using the trace elements of sphalerite (Fig. 12). In these diagrams, sphalerite from the Pb–Zn orebody in the Huangshaping deposit falls into the hydrothermal deposit areas and is different from skarn and MVT Pb–Zn deposits.

Previous studies on the isotopes of the Huangshaping deposit (Xi et al., 2009; Yuan et al., 2014; Ding et al., 2016) have revealed that the

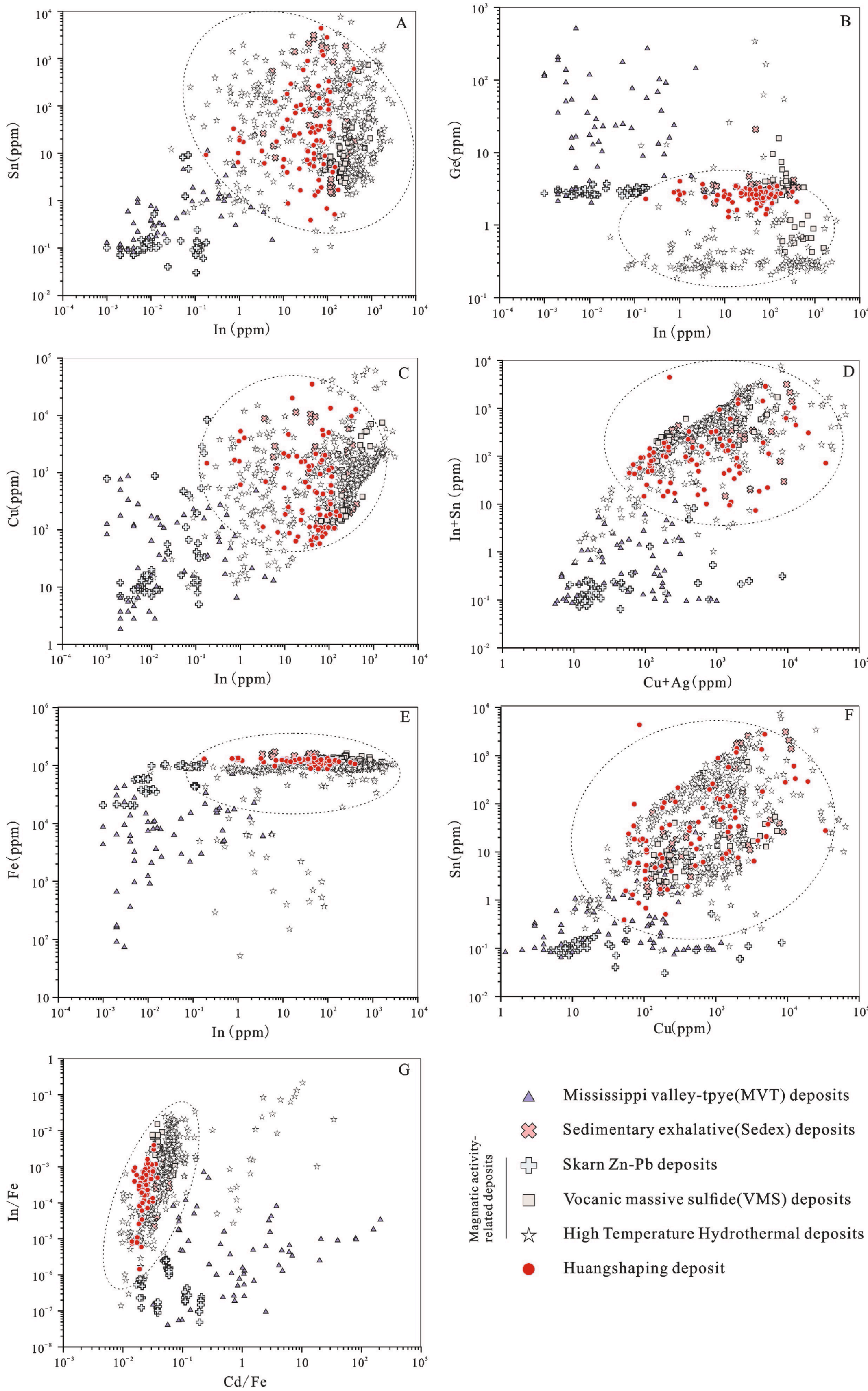


**Fig. 12.** Binary plots of (A) In vs. Sn, (B) In vs. Ge, (C) In vs. Cu, (D) Cu + Ag vs. In + Sn, (E) In vs. Fe, (F) Cu vs. Sn, and (G) Cd/Fe vs. In/Fe in sphalerite from the Huangshaping deposit and MVT, VMS, high-temperature hydrothermal, and skarn deposits. Data from Bauer et al. (2018); Cook et al. (2009); Hu et al. (2020); Ye et al. (2011).

$\delta^{34}S_{CDT}$‰ of sulfide is 4.00–16.3, whereas the $\delta^{34}S_{CDT}$‰ of magmatic rocks and wall rocks shows evident variation from 3.90 to 16.5 and −22.6 to −2.50, respectively. The $\delta^{34}S$ of the sulfides is close to that of the magmatic rocks, indicating that the genesis of the mineralization is related to the magmatic rocks. The initial $^{87}Sr/^{86}Sr$ ratio of sulfides is between that of the magmatic rock and wall rocks, indicating that the ore-forming fluid resulted from the mixing of magmatic hydrothermal fluid and formation water. The Pb isotopes show that the ore-forming material and ore-forming fluid are of crust–mantle mixed origin, and the metallic elements may have originated in the ancient lower crust (Li et al., 2019a; Li et al., 2019b). Moreover, the ore-related magma of this deposit experienced enrichment in metallic elements and sulfur at depth (Ding et al., 2016; Hu et al., 2017; Ding et al., 2021).

Based on the evidence, the Pb–Zn mineralization of the Huangshaping copper-polymetallic deposit is formed by the residual magmatic hydrothermal fluid. The hydrothermal fluid associated with Pb–Zn mineralization evolved from the hydrothermal fluid associated with W–Mo(Fe) mineralization that migrated and cooled along the structure over long distances.

## 6. Conclusions

(1) Sphalerite enriches Ga, In, Cd, Mn, Cu, Ag, Sn, and Pb; pyrite enriches Tl, Sb, W, and Se; and pyrrhotite enriches Ge. Sphalerite is the main carrier mineral of dispersed (Ga, In, and Cd) and ore-forming elements (Fe, Cu, Mn, Ag, and Sn).

(2) In sphalerite, Fe, Mn, Cd, Ga, In, Cu, Sn, and Ag exist by substituting for other elements (such as Zn). The sphalerite has been replaced by late-stage galena, and Sb exists in galena. The replacements for Cu, Sn, Ga, Ge, and In in sphalerite likely enter into the lattice as the following substitutions: $mCu^{2+}+nSn^{2+}\leftrightarrow(m+n)Zn^{2+}$, $Ge^{4+}+2Cu^{+}\leftrightarrow3Zn^{2+}$, $In^{3+}(Ga^{3+})+Cu^{+}\leftrightarrow2Zn^{2+}$. In pyrite, W occurs as solid solution. In pyrrhotite, Ge is incorporated through coupling substitution. In pyrite and pyrrhotite, Pb exists in the late-stage mineral assemblage (e.g., galena) which replaces the pyrite and pyrrhotite, whereas Ag, Se, Tl, Sb, and other elements exist in late-stage minerals (e.g., galena).

(3) Sphalerite, which formed during the Pb–Zn mineralization stage of the Huangshaping copper-polymetallic deposit, forms in a medium–high-temperature environment (288–341 °C). The Pb–Zn mineralization is the result of the residual magmatic hydrothermal fluid, which is related to skarn carrying the ore-forming elements in the intrusion, transporting to distal, and finally leading to Pb–Zn mineralization. The Pb–Zn mineralization, along with the W–Mo(Fe) mineralization, constitutes a skarn mineralization system.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.oregeorev.2022.104867.

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